

Muon's decay time and parity symmetry breaking



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Abstract

The measurement of the muon lifetime and decay dynamics provides fundamental experimental evidence for Special Relativity and the Weak Interaction. This experiment aims to observe muon decay and demonstrate parity symmetry violation in weak decays. A modular experimental setup was designed, tested, and assembled, consisting of three plastic scintillators coupled to Photomultiplier Tubes (PMTs), a modular electronic signal-processing chain, a digital data acquisition system (digitizer), and a solenoid producing a uniform magnetic field. To isolate valid muon decay events, a coincidence/veto trigger system was implemented to detect the energy deposited by decay electrons within the scintillators in different materials (air and salt). The setup successfully demonstrated parity violation by observing the asymmetric angular distribution of decay electrons and spin precession of decaying muons as a function of the applied magnetic field.

Everything that is talked about is accessible on GitHub [1]

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1 Theoretical Introduction

1.1 Cosmic Rays and Atmospheric Muons

Primary Cosmic Rays are high energy astroparticles generated from outside of the Solar System and from Sun. When they enter the atmosphere, they interact with molecules in air, producing shower of secondary particles. Following interaction and decay processes, some particles reach the Earth surface.

Atmospheric muons are the most abundant charged particles reaching the surface. This is due to the small energy loss in the atmosphere, the small interaction cross section and a long lifetime, since its decay is mediated by weak interactions.

Atmospheric muons are mainly produced by the decay of secondary charged particles. The two main channels of production, with their branching ratios [2], are:

$$\begin{aligned}\pi^\pm &\rightarrow \mu^\pm \nu_\mu^{(-)} \quad B.R. : (99.98770 \pm 0.00004)\% \\ K^\pm &\rightarrow \mu^\pm \nu_\mu^{(-)} \quad B.R. : (63.56 \pm 0.11)\%\end{aligned}\tag{1.1}$$

1.2 Stopping power

When passing through matter, muons lose energy along their path inside the material. The average energy loss due to the interaction with matter, as a function of the energy of the muon, follows the expression:

$$\frac{dE}{dx} = a(E) + b(E) \times E\tag{1.2}$$

where $a(E)$ is the contribution due to ionization and atomic excitation of the material and $b(E) \times E$ is the contribution due to bremsstrahlung, pair production, photonuclear emission and radiative loss.

In Figure 1 are shown the stopping power curves by varying the energy of the muon, for 3 materials that will be used in this experiment and air. Polyvinyltoluene is the material of the detector used, while sodium chloride and aluminum will be used for specific measurements. Data are obtained from PDG [2].

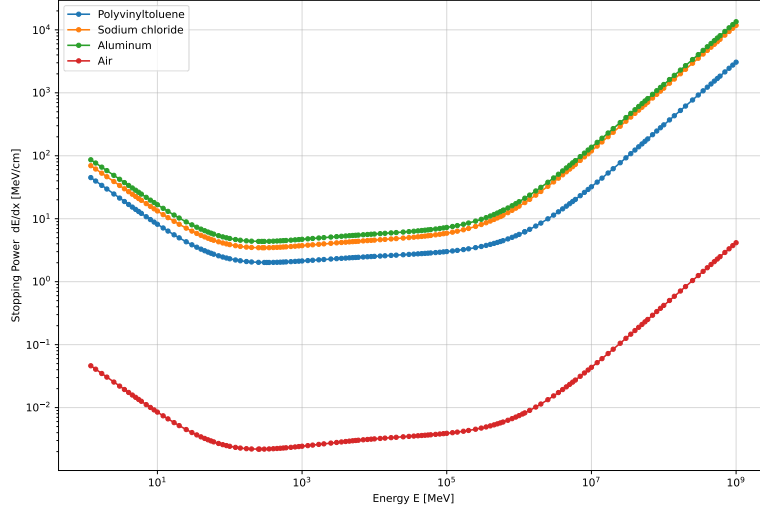


Figure 1: Stopping power, as a function of the muon energy, for different materials

By integrating the inverse of the stopping power we can find the maximum range of muons through every material, using Eq. (1.3). In Figure 2 is shown this relation as a function of energy.

$$\Delta x(E) = \int_0^x dx = \int_0^E \frac{dx}{dE} dE \quad (1.3)$$

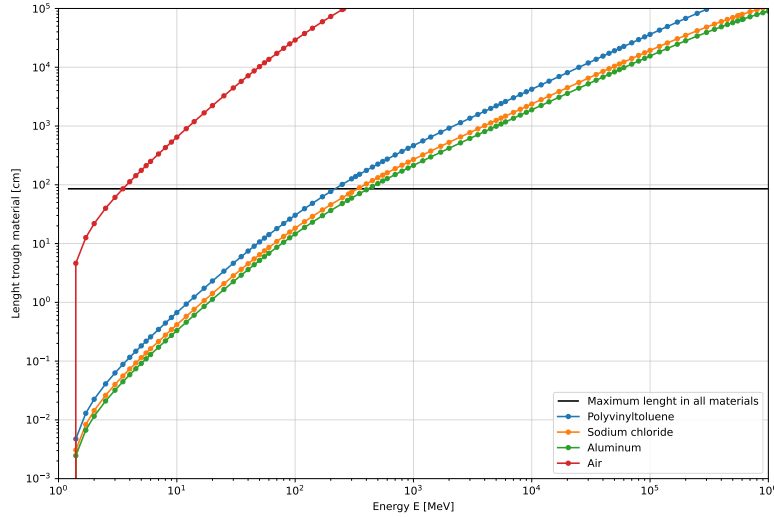


Figure 2: Length through matter for different material, and range in every material

The horizontal line indicates the longest straight path a particle can perform in each material, taken as the diagonal of the smallest scintillator in our setup (85.45 cm). For Aluminum and Sodium Chloride we used blocks with surface similar to the one of the

scintillator, and some centimeter different in heigh, causing to have a maximum length difference of the order 10^{-1} cm. Therefore only one line is showed, since in logarithmic scale the difference is visually negligible.

From Figure 2 we can extrapolate the energy range of stopped muons in the materials shown, taking all the energies for which the maximum length is under the line drawn in the graph.

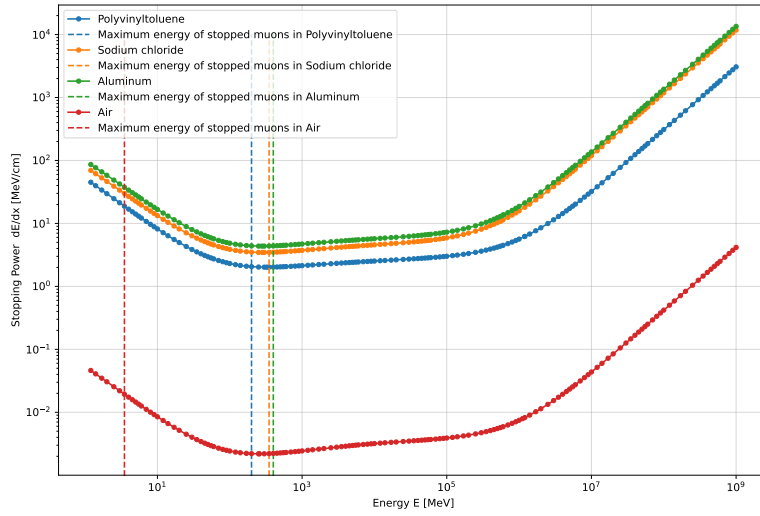
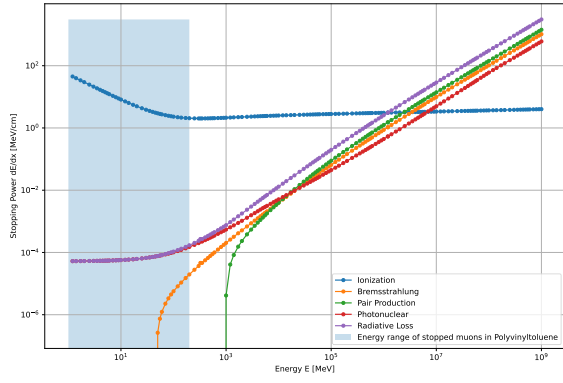
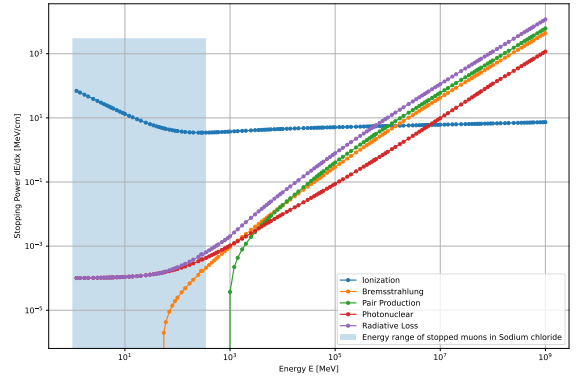


Figure 3: Stopping power and energy ranges, as a function of the muon energy, for different materials. In Figure 3 we can see the energy ranges thus obtained on the stopping power graph. This clearly shows that, for similar lengths, air adds a negligible contribute, since it has a smaller energies interval that can stop muons (with the same path length) and its stopping power is around 2 orders of magnitude lower than the other materials considered.

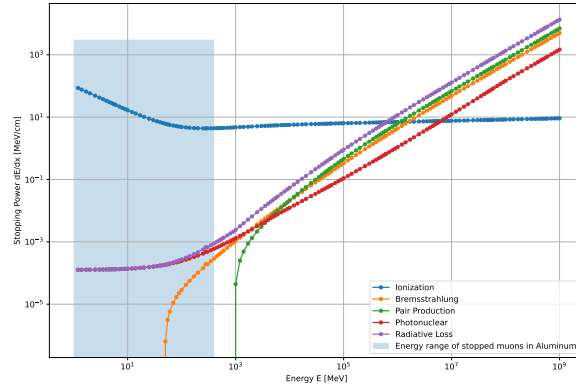
We can now use this estimated energy range to also see the different contributions of the stopping power we enlisted before. In Figure 4 are showed all the different contributions for the 3 materials we are interested in.



(a) Polyvinyltoluene



(b) Sodium chloride



(c) Aluminum

Figure 4: Stopping power contributions for different materials

We can notice how, at low energies, the contribution of ionization is at least 4 orders of magnitude more intense than the other contributions. Therefore we can assume that, for the energy range of our interest, only the ionization contribution can be considered.

1.3 Properties of Atmospheric Muons

Since the interaction with matter of μ^- and μ^+ is different, it is important to know the so called “muon charge ratio”, defined as the ratio of positive and negative atmospheric muons. Different measures showed a muon charge ratio varying with the energy (as shown in [3]). For low energy muons (≈ 1 GeV) the muon charge ratio $R_\mu = 1.18 \pm 0.02$ [4].

This gives a the probabilities to have a μ^+ and μ^- :

$$P_+ = (54.1 \pm 0.4)\%$$

$$P_- = (45.9 \pm 0.4)\%$$

Atmospheric muons' angular distribution depends on their energy, as shown in Figure 5.

Muons with low energies (in which we are interested, as shown in the section Sec. 1.2) follow a $\cos^n \theta$ distribution, with $n \approx \frac{2}{3}$.

We are also interested in the muon flux at sea level for low energy particles. In Figure 6 is reported the graph of the integral (in $cm^{-2}sr^{-1}s^{-1}$) and differential (in $cm^{-2}sr^{-1}s^{-1}(\frac{GeV}{c})^{-1}$) muon spectra, as reported by *Rastin* [5].

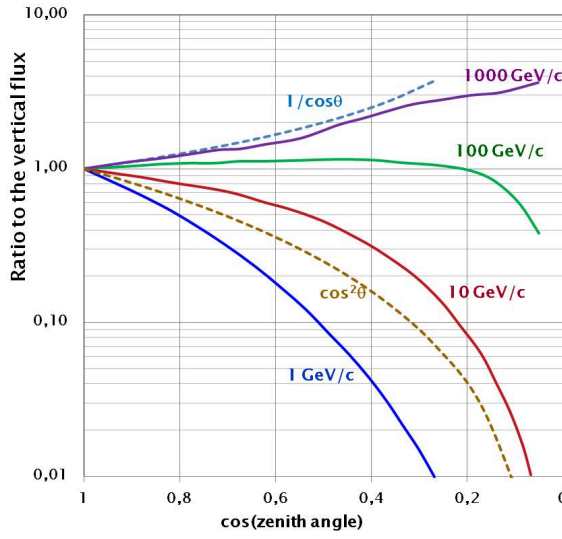


Figure 5: Angular distributions of atmospheric muon at different energies [6]

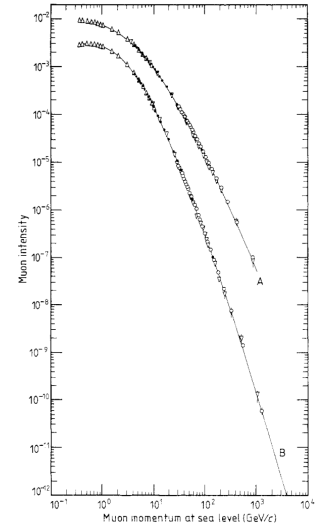


Figure 6: Integral (A) and differential (B) muon spectra [5]

1.4 Muons decay

Muon decays mainly ($\approx 100\%$) into an electron and two “neutrinos” ($\nu_\mu + \bar{\nu}_e$ for μ^- , and $\bar{\nu}_\mu + \nu_e$ for μ^+), as shown in Figure 7:

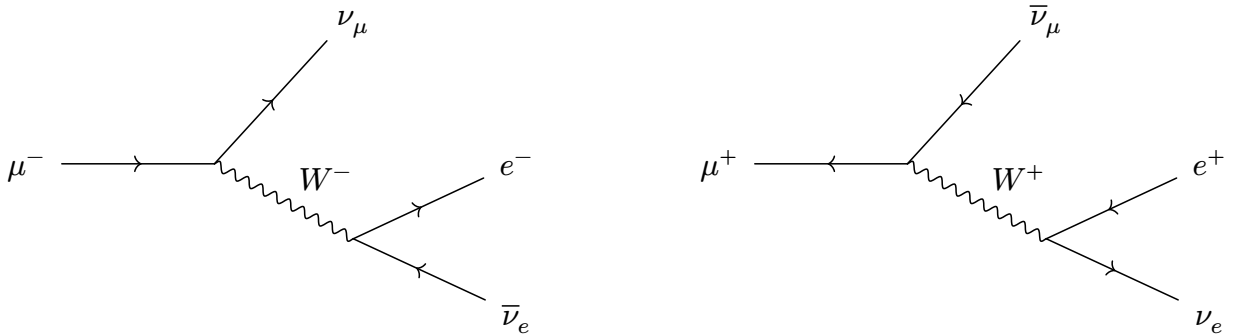


Figure 7: Feynman diagrams of muon's β decays

The other possible decays are higher order diagrams of this β decay, those of which imply a photon emission [2]. This means that, in order to measure muon lifetime, only a time measurement of the produced electron is needed.

Muon spontaneous decay time distribution follows an exponential, with a mean life $\tau = (2.1969811 \pm 0.0000022)\mu s$ (as listed in PDG [2]). Due to the exponential distribution, the time in which the μ is generated is not needed, since every time offset can be reabsorbed in the coefficient of the distribution. Hence, also atmospheric muons can be used to measure μ mean lifetime.

1.5 μ^- interaction with matter

When a muon stops in matter, it is subject to electromagnetic potential of the nuclei. In this case, before decaying, the muon can be captured by an atom. Since it's not an electron, the captured muon is not subject to the Pauli exclusion principle, and it occupies the innermost orbital. Since the muon is more massive than the electron, his $1s$ orbital is less wide than the electron's one. Hence, the PDF of the muon has a non-negligible overlap with the protons in the nuclei, causing an interaction among them. The tree-level diagram of the interaction between the muon and the proton is showed in Figure 8.

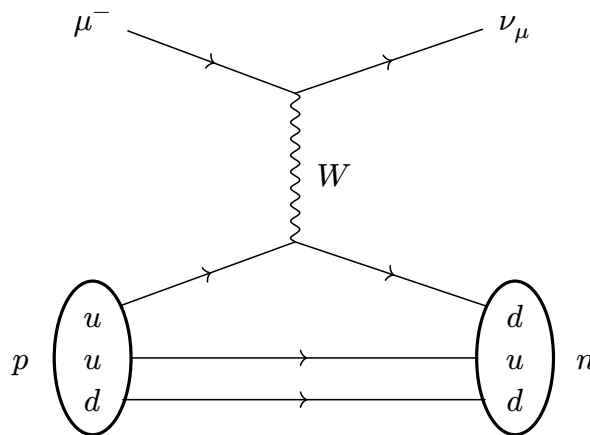


Figure 8: Feynman diagrams of $\mu^- p$ interaction

In this process, the muon rest energy gets redistributed among the neutrino and the neutron. Due to the neutrino is non-interacting, we will not be able to see it, so we have to search for some evidence of the excitation of the neutron, since it will gain

some energy from the muon mass. The excited neutron goes back to its fundamental state through different paths: it can simply emit a photon or it could lose its energy through internal conversion, emitting one of the orbital electrons of the atom. Since photons from radiative emission have energies comparable with ambient photons, that we already exclude because they form a disturbing background, we will not be able to detect them.

The capture rate of the muon will depend from the intensity of the EM interaction with the nuclei, so will depend on the Z_{eff} , as discussed by *Primakoff* [7]. The simplified Primakoff formula is reported in Eq. (1.4)

$$\Lambda_{\text{cap}}(A, Z) = Z_{\text{eff}}^4 X_1 \left[1 - X_2 \left[\frac{A - Z}{2A} \right] \right], \quad (1.4)$$

where X_1 is related the muon capture rate in hydrogen, X_2 is a factor that consider the Pauli exclusion principle in the nuclei, and Z_{eff} is obtained as discussed by *Ford and Willis* [8].

We are interested in the capture rate and in the mean life of captured muons inside Aluminum and Sodium Chloride. In advance, we should also consider the elements inside the scintillator as a possible nuclei capable of capturing the muon. But, as reported by *Suzuki et al.* [9], in both Hydrogen and Carbon μ^- has lifetimes similar to the vacuum one, with rates at least one order of magnitude smaller, that we are not able to distinguish. In Table 1 are reported those data from *Suzuki et al.*[9], and the Z_{eff} .

Element	Z_{eff}	$\tau(\text{ns})$	R (MHz)
Al	11.48	864 ± 2	0.662 ± 0.003
Na	9.95	1204 ± 2	0.3772 ± 0.0014
Cl	14.24	560.8 ± 2.0	1.333 ± 0.006

Table 1: Muon capture data for different elements, obtained from [9].

In order to obtain the decay time of captured muon in Sodium Chloride, we should take the weighted mean of the τ of Na and Cl, taking into account the relative capture rate of those two elements.

$$\tau_{\text{NaCl}} = \frac{\sum_{i=1}^2 R_i \tau_i}{\sum_{i=1}^2 R_i} \quad \sigma_{\tau_{\text{NaCl}}} = \sqrt{\sum_{i=1}^2 \left[\left(\frac{d\tau_{\text{NaCl}}}{dR_i} \sigma_{R_i} \right)^2 + \left(\frac{d\tau_{\text{NaCl}}}{d\tau_i} \sigma_{\tau_i} \right)^2 \right]}$$

with $i = 1, 2$ indicating, respectively, Na and Cl. By performing those calculations, we obtain

$$\tau_{\text{NaCl}} = (702.7 \pm 1.7) \text{ ns.} \quad (1.5)$$

Meanwhile, to obtain the total decay rate of captured muons in Sodium Chloride, we take the mean of the two rates.

$$R_{\text{NaCl}} = \frac{\sum_{i=1}^2 R_i}{2} \quad \sigma_{R_{\text{NaCl}}} = \sqrt{\sum_{i=1}^2 \left(\frac{dR_{\text{NaCl}}}{dR_i} \sigma_{R_i} \right)^2}$$

with $i = 1, 2$ indicating, respectively, Na and Cl. By performing those calculations, we obtain

$$R_{\text{NaCl}} = (0.855 \pm 0.003) \text{ MHz.} \quad (1.6)$$

Using those rates, we can now obtain the rate of muons, in each material, that decays freely or gets captured. The rate of the free decay is given by the τ of the muon.

$$R_{\text{free}} = \Gamma_{\text{tot}} = \frac{1}{\tau_{\mu}} \quad \sigma_{R_{\text{free}}} = \frac{\sigma_{\tau}}{\tau_{\mu}^2}$$

Using $\tau_{\mu} = (2.1969811 \pm 0.0000022) \mu\text{s}$ we obtain

$$R_{\text{free}} = (0.4551701 \pm 0.0000005) \text{ MHz.} \quad (1.7)$$

In a given material with rate of capture R_{mat} , we can compute the percentage of muons that decay freely and through capture. We only consider the error on R_{mat} since the error on R_{free} is various order of magnitude lower than the σ of the materials we are considering for this experiment.

$$R_{\text{free}}^{\%} = \frac{R_{\text{free}}}{R_{\text{free}} + R_{\text{mat}}}, \quad R_{\text{mat}}^{\%} = \frac{R_{\text{mat}}}{R_{\text{free}} + R_{\text{mat}}}, \quad \sigma_{R^{\%}} = \frac{R_{\text{free}} \sigma_{R_{\text{mat}}}}{(R_{\text{free}} + R_{\text{mat}})^2}.$$

Material	$R_{\text{free}}^{\%}$	$R_{\text{mat}}^{\%}$	$\sigma_{R^{\%}}$
Al	40.7%	59.3%	0.1%
NaCl	34.74%	65.26%	0.08%

Table 2: Probability that a muon decays freely or through capture in different materials.

1.6 μ^+ interaction with matter

When a positive muon gets thermal inside a material, it can capture an electron to form the muonium $M = \mu^+ e^-$. Due to the instability of μ^+ the muonium decays as shown in Figure 9.

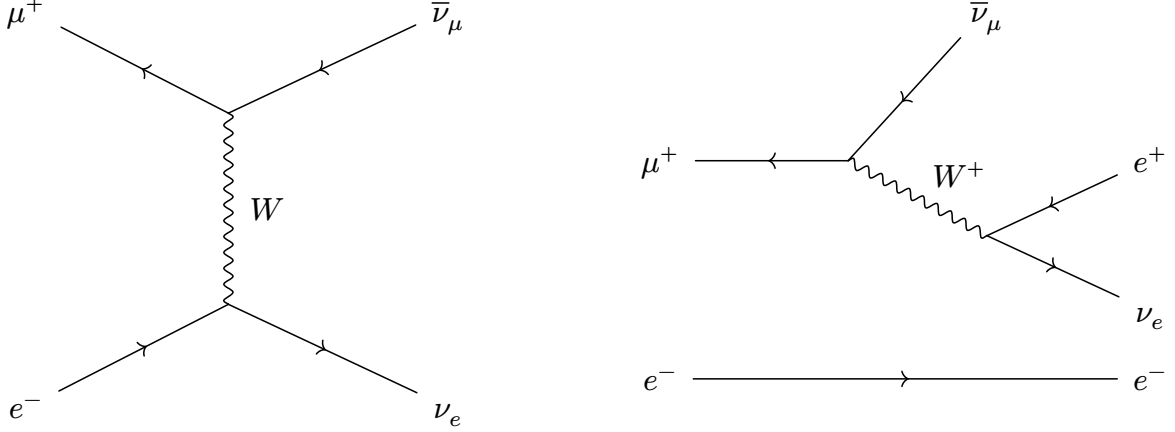


Figure 9: Feynman diagrams of muonium decays

Although the invisible decay still hasn't been observed, Standard Model predictions suggest the process would have a branching ratio $\text{Br}(M \rightarrow \nu_e \bar{\nu}_\mu) = 6.6 \times 10^{-12}$ [10]. The muonium preferably ($\approx 100\%$) decays into $e^+ e^- \nu_e \bar{\nu}_\mu$. The decay time is similar to the free anti-muon decay, but gets some corrections due to the interaction of the two particles. As discussed by *Czarnecki et al.* [11], the leading effects, caused by the motion of the μ^+ in the rest frame of the muonium, generate a correction term

$$\tau_M = \tau_\mu \left(1 + \frac{\alpha^2 m_e^2}{2m_\mu^2} \right), \quad \left(\frac{\alpha^2 m_e^2}{2m_\mu^2} \right) \approx 6 \times 10^{-10}.$$

Since the correction is really small, we will not consider it, and simply take $\tau_M = \tau_\mu$.

1.7 Up-Down Asymmetry

Atmospheric muons are mainly produced from a two body decay of charged kaons and pions, as already mentioned in Sec. 1.1. Both those mesons are spinless particles, so the final state has to be a singlet state of spin. In the meson's rest frame, this means that the muon and the neutrino must have same helicity (opposite spin direction). In SM, neutrino are massless, so ν must be left-handed and $\bar{\nu}$ must be right-handed. Therefore μ^- must be produced right-handed, while μ^+ must be produced left-handed. But, since

π and k are highly energetic, the products of the decay will receive a Lorentz boost, and helicity is not Lorentz invariant. This means that is possible to see at sea level some muons with the “wrong” polarization.

Due to the V-A structure of the weak coupling, there is a correlation between the direction of the produced $e^\pm$ and the spin orientation of the $\mu^\pm$, with a differential decay width:

$$\frac{d^2\Gamma_\pm}{dx d\Omega} = \frac{G_F^2 m_\mu^5}{192\pi^3} x^2 \left[(3 - 2x) \pm P_\mu (2x - 1) \cos \theta \right] \quad (1.8)$$

with $G_F \approx (1.1663787 \pm 0.0000006) \times 10^{-5} \text{ GeV}^{-2}$ Fermi coupling, $m_\mu \approx (105.6583755 \pm 0.0000023) \text{ MeV}$ mass of the muon, $x = \frac{2E_e}{m_\mu}$ energy fraction of the $e^\pm$, θ angle between $\mu^\pm$ spin direction and direction of the $e^\pm$ momentum and P_μ polarization of the $\mu^\pm$, that, for an energy range of cosmic muons $E = (0.75 - 1.17) \text{ GeV}$, is $P_\mu = -0.33 \pm 0.04$ [12]. By integrating over $x = [0, 1]$ to consider all possible energy fractions, we obtain:

$$\frac{d\Gamma_\pm}{d\cos\theta} = \frac{\Gamma_\mu}{2} [1 \pm A' \cos \theta] \quad (1.9)$$

with $A' = \frac{1}{3} P_\mu \xi$ integral asymmetry [13], and $\xi = 1$ is one of the Michel parameters, that describe the phase space distribution of charged leptons decays into lighter leptons. Now we can integrate over θ , in order to separate the upward decays and the downward ones. Since, in the lab-frame, μ^- are mainly right-handed, the electron goes upward for $\theta = [\pi, \frac{\pi}{2}]$. Meanwhile, since μ^+ are mainly left-handed, the positron goes upward for $\theta = [0, \frac{\pi}{2}]$. This means that upward $U(t)$ and downward $D(t)$ decays will follow these relations

$$U(t) = U_0 \left(1 + \frac{A'}{2} \right) e^{-\frac{t}{\tau_\mu}} \quad (1.10)$$

$$D(t) = D_0 \left(1 - \frac{A'}{2} \right) e^{-\frac{t}{\tau_\mu}} \quad (1.11)$$

obtained assuming no depolarization effects or background events.

Now we can define the up-down asymmetry as

$$A = \frac{U(t) - D(t)}{U(t) + D(t)} = \frac{(U_0 - D_0) + \frac{A'}{2}(U_0 + D_0)}{(U_0 + D_0) + \frac{A'}{2}(U_0 - D_0)} \quad (1.12)$$

If $U_0 = D_0$, we would obtain $A = \frac{A'}{2}$. We can also notice that $\frac{A'}{2} = \frac{1}{6}P_\mu\xi = -0.055 \pm 0.007$ is a really small factor. Therefore, we are able to simplify Eq. (1.12), grouping $(U_0 + D_0)$, obtaining

$$A = \frac{(U_0 - D_0) + \frac{A'}{2}(U_0 + D_0)}{(U_0 + D_0) + \frac{A'}{2}(U_0 - D_0)} = \frac{\frac{U_0 - D_0}{U_0 + D_0} + \frac{A'}{2}}{1 + \frac{A'}{2} \frac{U_0 - D_0}{U_0 + D_0}} \approx \frac{U_0 - D_0}{U_0 + D_0} + \frac{A'}{2} \quad (1.13)$$

In order to measure this asymmetry, one could put a magnetic field B with a component orthogonal to the spin direction. By doing so, the spin will precess around the magnetic field axis, changing its direction. This process follows the Larmor precession, which tells us the spin will precess around the B direction with a frequency:

$$\omega_\mu = -\frac{egB}{2m_\mu} \quad (1.14)$$

with e electric charge and $g \approx 2$ gyromagnetic factor (neglecting corrections above tree-level). This effect makes the polarization time dependent with a periodic law, obtaining $A'(t) = \frac{1}{3}P_\mu\xi \cos(\omega_\mu t)$. This means that the total asymmetry with magnetic field will be

$$A_{field} \approx \frac{U_0 - D_0}{U_0 + D_0} + \frac{A'(t)}{2} \quad (1.15)$$

Taking the difference of the 2 cases (with and without field) isolates the time dependent term, giving

$$A_{field} - A = \frac{A'(t)}{2} - \frac{A'}{2} = \frac{1}{6}P_\mu\xi \left(\cos(\omega_\mu t) - 1 \right) \quad (1.16)$$

2 Instrumentation and Modular Electronics

2.1 Experimental Methodology and Workflow Architecture

In order to transform raw physical interactions into reliable datasets we used the so-called “modular” approach: all acquisition process is mediated by many different components and each of them has its own purpose. So, rather than treating an acquisition setup as a rigid single instrument, we built our setup upon standard modular electronics that we arranged together.

This “modular” approach requires 4 steps to be able to collect reliable data:

1. **Design and Functional Assembly:** Conceptualizing the signal path logic and physically staging the individual hardware modules within the crate. This is influenced by the available and functioning modules we had access to and our choices of what was more important for us (as will be discussed in Sec. 3).
2. **Component Characterization:** Separately testing and validating each hardware component to check if each of them works as we would expect.
3. **System Calibration and Integration:** Combining the different modules into an interconnected, synchronized chain. Tuning thresholds, time delays, cables, exc. so that an initial signal can go through all the chain and give a resulting signal. This step also verify what in step 1. was theorized.
4. **Data Acquisition:** Launching the data collection and digitization for offline physical analysis.

This approach is very efficient in pursuing troubleshooting for each module that can lead to instrument dependencies and electronic anomalies, which can be systematically identify and correct.

2.2 Experimental Infrastructure and Auxiliary Equipment

The experimental setup follows the **Nuclear Instrumentation Modules (NIM)** standard, using a dedicated crate to house and power the modular electronics. This crate provides the essential high-stability power rails (± 6 V, ± 12 V, ± 24 V) and ensures a common grounding scheme. This configuration allows the minimization of the electronic noise by preventing ground loops.

The following precision laboratory instruments have been used to test all the electronic modules to obtain a stable and reliable acquisition:

- **Tabor 8020 Pulse Generator (20 MHz)** served as the primary signal source for module characterization, providing reproducible pulses with adjustable parameters to simulate real detector events.
- **Tektronix TDS 2014B Oscilloscope (100 MHz, 1 GS/s)**
- **Multimeter Fluke 115 True RMS**

2.3 Data Acquisition System and NIM Modules

The signal processing chain is designed to transform analog pulses into calibrated logic signals for real-time coincidence in order then to be processed and digitalized. The workflow is built by a suite of NIM-standard modules, which handle everything from signal replication to temporal logic. The instrumentation used is organized as follows:

- **Discriminator Mod. N 96 (CAEN):** Converts negative analog signals into standardized NIM logic pulses based on a tunable voltage threshold, serving as the primary trigger source.
- **Octal Fan-Out Mod. 430 (LeCroy):** An 8-channel distribution module used to replicate logic signals across various branches of the acquisition chain.
- **Logic Unit Mod. N405 (CAEN):** A programmable 4-channel unit used to perform Boolean operations (AND, OR, XOR).
- **Dual Timer N93B (CAEN):** This module generates logic gate signals of adjustable duration upon receiving an input trigger, allowing for precise control over the coincidence window and gate width used in the acquisition chain.
- **Dual Delay Module:** This module provides accurate fine-tuning of the temporal alignment of signals, compensating for differences in cable lengths and propagation delays to ensure perfect synchronization between different channels.
- **Counter Mod. N1145 (CAEN):** Used for real-time monitoring of event rates and total counts, providing an immediate visual check of the experimental flux. Mostly used during the “testing” part of the experiment.
- **Digitizer DT5720B (CAEN):** A 12-bit ADC with a 250 MS/s sampling rate, performing digital charge integration and waveform reconstruction for offline analysis.
- **Power Systems:** A Programmable Power Supply and a High Voltage (HV) Power Supply providing the stable bias voltages required for the detectors and the high-stability current for the NIM crate rails.

2.4 Logic Verification and Signal Distribution

The acquisition chain was initially tested to ensure the correct distribution of signals and proper synchronization across all timing branches. During this phase we proceeded in testing and characterizing each of the modules in order to ensure the proper functioning and performance. The first module we tested was **LeCroy 430 Octal Fan-Out**: we fed the logic NIM signals into the module to characterize it and to verify its preservation of fast rise times (< 2 ns).

During the testing/troubleshooting phase we have tested all of the channels of the module. The results of the test have, as shown in Figure 10, a constant delay of 5 ns overall channels and a very low distortion of the input signal.

To test the counter **CAEN Mod. N1145** we used the following technique: knowing the frequency of a periodic signal generated by the pulse generator and setting a fixed acquisition time, it was possible to check the reliability of the counts displayed by the counter. All the channels of this instrument showed excellent behavior.

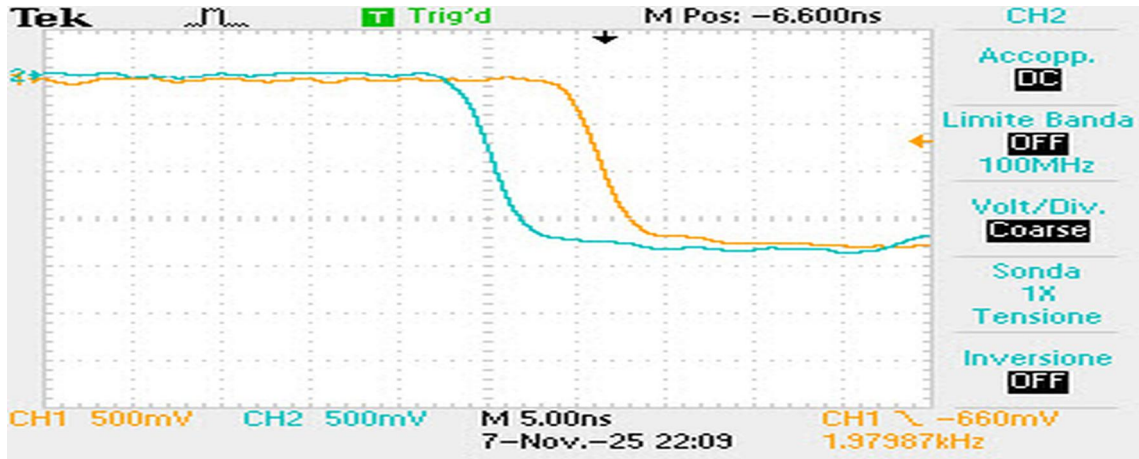


Figure 10: Oscilloscope trace testing Fan-Out module.

2.5 Characterization of the Discriminator

Mod. N96

Prior to the selection of the current module, the **CAEN Mod. N417** was initially considered; however, it resulted unsuitable for the experimental requirements due to significant hardware degradation and stability issues. In trying to test the module, several channels resulted damaged, including compromised adjustment screws that hindered precise calibration, and the threshold levels proved to be highly unstable over time. In addition, the N417 module failed to meet the dynamic range requirements of the experiment, as its maximum threshold was insufficient compared to the 250 mV signals targeted for rejection. Therefore, the **CAEN N96** was adopted to provide the necessary mechanical reliability and threshold stability.

A fundamental stage of the characterization for this module involved the linearity calibration of its central channels (CH4 to CH7, the ones we used). In fact, although

all the channels have been tested, only the channels 4 through 7 were reliable because the other channels we observed several problems, such as compromised adjustments screws and stability issues. This process was essential to establish the exact relationship between the manual potentiometer settings (V_{th}) and the actual minimum input amplitude (V_{in}) required to fire the discriminator. The experimental setup used the **Tabor 8020** to generate negative square pulses, which were split via a Fan-Out: one branch was sent to the oscilloscope for amplitude monitoring, while the other was fed into the **N96** discriminator. Thresholds were swept from -50 mV to -225 mV and measured with a precision multimeter. To account for mechanical hysteresis and ensure consistency, a “unidirectional approach” was adopted: we set each target threshold reaching it from a lower value to an higher one.

The resulting data were modeled using the following linear regression:

$$V_{in} = a \cdot V_{th} + b \quad (2.1)$$

In order to estimate the errors, we measured the oscillation of the set threshold.

We measured an oscillation of $\Delta V_{th} \approx 2$ mV for the channels 4, 5, 7 and a $\Delta V_{th} \approx 3$ mV for channel 6, therefore for the statistical analysis we have modeled the error with an uniform distribution in which $\sigma = 3$ mV is applied for channel 6 and $\sigma = 2$ mV for the remaining channels.

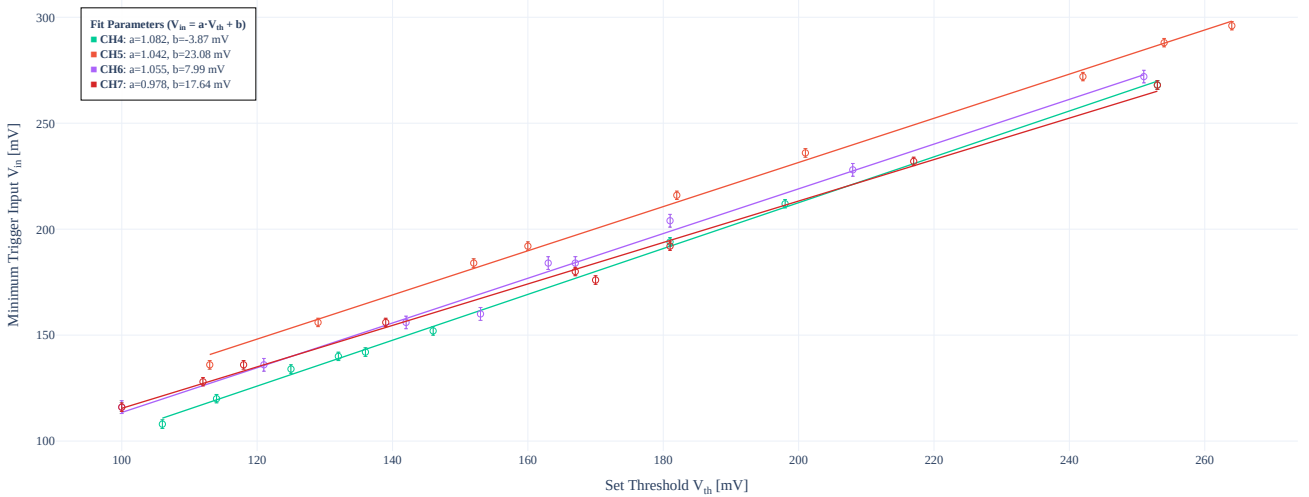


Figure 11: Linear regression of the N96 Discriminator threshold calibration for channels CH4-CH7.

To evaluate the goodness-of-fit beyond the reduced Chi-squared ($\sim \chi^2_\nu$), the corresponding p -values were computed using the cumulative Chi-squared distribution function with

$\nu = N - k$ degrees of freedom, where N is the number of experimental data points and $k = 2$ represents the free parameters of the linear regression (slope and intercept).

Channel	a (Slope)	σ_a	b (Intercept)	σ_b	$\tilde{\chi}_\nu^2$	p -value
CH4	1.110	0.023	-7.66	3.36	1.53	0.18
CH5	1.042	0.021	23.08	4.09	2.14	0.07
CH6	1.055	0.053	7.99	8.98	1.70	0.13
CH7	0.978	0.026	17.64	4.43	2.01	0.09
<i>Note: All voltage measurements (b, σ_b) are expressed in mV. Input voltage uncertainty $\sigma_{V_{in}} = 2$ mV for CH4, CH5, CH7 and $\sigma_{V_{in}} = 3$ mV for CH6.</i>						

Table 3: Fit parameters, uncertainties, reduced Chi-squared, and p -values for the N96 discriminator channels.

To evaluate the goodness-of-fit beyond the reduced Chi-squared ($\sim \chi_\nu^2$), the corresponding p -values were computed using the cumulative Chi-squared distribution function with $\nu = N - k$ degrees of freedom, where N is the number of experimental data points and $k = 2$ represents the free parameters of the linear regression (slope and intercept).

As reported in Table 3, all discriminator channels exhibit reduced Chi-squared values close to unity, ranging from 1.53 to 2.14. The associated p -values remain above the standard statistical significance threshold ($\alpha = 0.05$), confirming that the linear model properly describes the response of the N96 discriminator and that the assigned experimental uncertainties $\sigma_{V_{in}}$ are realistic and well-estimated.

The p -value represents the probability of obtaining a Chi-squared statistic at least as extreme as the observed one, under the null hypothesis that the linear response model is correct. Since all calculated p -values exceed 0.05, there is no statistical evidence to reject the null hypothesis, demonstrating good agreement between the linear calibration model and the experimental measurements across all active channels.

The analysis revealed slopes (a) near unity, with slight deviations (0.978 to 1.110) reflecting internal gain factors specific to each channel. These values, alongside the intercepts (b), serve as calibration constants to accurately map physical thresholds during data acquisition.

Notably, CH4 exhibited non-linear saturation for thresholds exceeding 200 mV, causing the operational range to be strictly limited.

2.6 Characterization of the Dual Timer Mod. N93B

To verify the gate-generation capabilities and the temporal precision of the CAEN N93B Dual Timer, we performed an overlap analysis using two decoupled channels monitored via the oscilloscope. A master logic pulse from the generator was split: the first branch was routed directly to Channel 1 (oscilloscope) to act as the reference trigger, while the second branch was used to trigger the N93B Dual Timer, with its output monitored on Channel 2 (oscilloscope).

Both channels exhibit stable NIM logic levels transitioning from a baseline of 0 V down to approximately -0.85 V. The dual timer introduced a clean, programmable propagation delay of 18 ns relative to the reference signal's falling edge at $t = 0$ ns.

The testing of this module is important in order to maintain a good defined gate window in order to have reliability in the chain acquisition.

To map the response of the timer channels, a master trigger pulse was fed into the input, and the resulting output logic pulse durations were systematically measured across all coarse range selector positions using the oscilloscope. The calibration results for the four output channels (CH1–CH2) for each timer are summarized in Table 4.

Nominal Set- ting	Timer 1 (CH1)	Timer 1 (CH2)	Timer 2 (CH1)	Timer 2 (CH2)
50 ns	50 ns	50 ns	70 ns	3 ns
200 ns	138 ns	148 ns	192 ns	50 ns
1 μ s	720 ns	800 ns	1.9 μ s	60 ns
10 μ s	6.4 μ s	6.6 μ s	20 μ s	50 ns
100 μ s	24 μ s	30 μ s	163 μ s	50 ns
1 ms	56 μ s	76 μ s	1.5 ms	50 ns
10 ms	620 μ s	1.4 ms	15 ms	50 ns
100 ms	7 ms	8 ms	175 ms	50 ns
1 s	49 ms	127 ms	2 s	50 ns
10 s	—	—	—	50 ns
<i>Note: Measured output pulse widths correspond to discrete coarse knob selector positions on the CAEN N93B module.</i>				

Table 4: Measured output pulse widths for each coarse knob position on the CAEN Dual Timer Mod. N93B across all four output channels (CH1-CH4).

As detailed in Table 4, channels CH1, CH2 of timer 1 and CH1 of timer 2 scale monotonically with the rotary selector position, providing broad gate-width coverage from 50 ns up to full seconds. In contrast, Channel 2 (CH2) of the timer 2 remains fixed at a narrow, fast pulse width (~ 50 ns) across higher switch positions, effectively operating as an auxiliary prompt logic output. Therefore we have chosen to use only the timer 1 because it had both channel correctly functioning.

2.7 Characterization of the Logic Unit Mod.

N405

This module has been tested in order to verify that the minimum temporal synchronization window for a logic operation such as AND or OR had a fixed value to guarantee overlapping of signals. In this prospective a timing overlap test has been conducted.

The procedure involved generating identical negative NIM pulses, which were split and routed through physical passive delay stages to introduce a controlled relative time shift. These time-stamped logic pulses were then fed into the input channels of the N405

unit. By progressively adjusting the delay lines, the overlap width between the signals was incrementally varied while the logic unit’s output was continuously monitored.

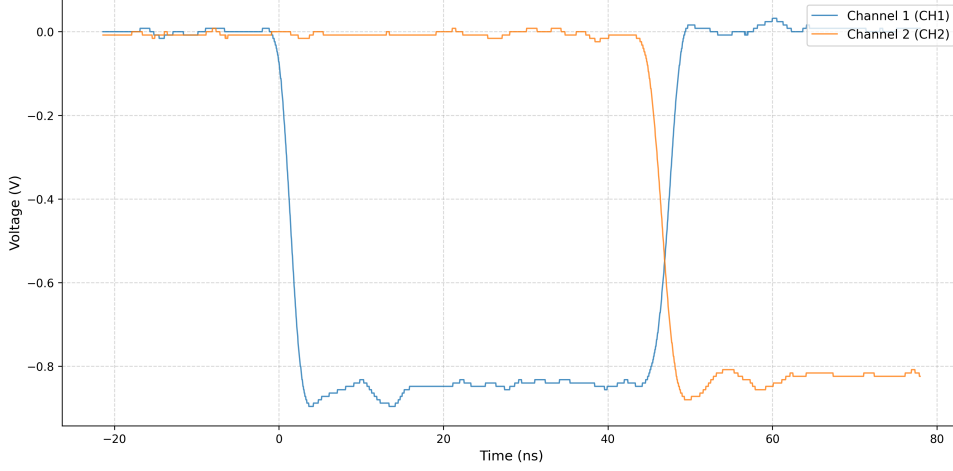


Figure 12: Oscilloscope screenshot displaying the coincidence window validation on the CAEN N405 Logic Unit.

Using the oscilloscope’s temporal cursors, the overlap between the falling edge of the trailing signal and the rising edge of the leading signal was measured to be exactly $\Delta t = 6$ ns. The test has been conducted in several stages and all confirmed that the Mod. N405 stably registers this 6 ns overlap without generating transient logic state anomalies or spurious “glitch” pulses. The edge case maximum are 6 ns, meaning that with a 50 ns signal window, a 6 ns of overlap and above results in accepting the event. This shape of the sharp edge temporal waveform in a very good index that the logic unit is capable of executing high-rate Boolean operations without drift, fulfilling a critical requirement for suppressing random background coincidences during the final data acquisition run.

2.8 Characterization of Dual Delay Module

To verify the precise passive phase-shifting capabilities of the Dual Delay module, we characterized it using the oscilloscope. The primary aim of this test was to accurately quantify the hardware-induced temporal shift and evaluate edge degradation or amplitude attenuation across the delay line.

We fed a master logic pulse into a Fan-In/Fan-Out module to duplicate the signal into two identical branches: a reference branch routed directly to Channel 1 of the

oscilloscope, and a second branch passed through the passive Dual Delay module under test before being monitored on Channel 4.

Direct observation on the oscilloscope confirmed that the set delay value matched the measured time shift. Furthermore, a side-by-side comparison of the waveforms verified that the input and output signals were similar in profile: the delayed pulse preserved its sharp, sub-nanosecond rise time and full NIM voltage amplitude without suffering from edge slurring, distortion, or attenuation. This transparent propagation confirms the module’s reliability for fine-tuning signal alignment across independent detection channels prior to coincidence logic processing.

2.9 Data Acquisition and Waveform Digitization

The final stage of the acquisition chain is the **CAEN DT5720B Digitizer**, a 12-bit ADC operating at a 250 MS/s sampling rate. The primary objective of the digitizer calibration was to establish a precise mapping between raw ADC counts and physical voltage levels (V). Given the 12-bit resolution distributed over a 2 Vpp dynamic range, the theoretical conversion factor is determined by:

$$\Delta V = \frac{2V}{2^{12}} \approx 0.488 \text{ mV/count} \quad (2.2)$$

Data acquisition and visualization parameters were managed via the CAENScope software interface, where the acquisition window was configured with a record length of 2048 sampling bins per event. At a clock speed of 250 MS/s (corresponding to a 4 ns time step per bin), this arrangement yielded a comprehensive observation window of 8.192 μs , providing more than sufficient temporal margin to capture the entire evolution of the pulse alongside its surrounding baseline.

Before starting with the data collection, the software has been used and tested in different configurations in order to understand its behavior in terms of noise and offsets.

3 Electrical setup

The setup we chose to minimise dead time and pileup probability is the following:

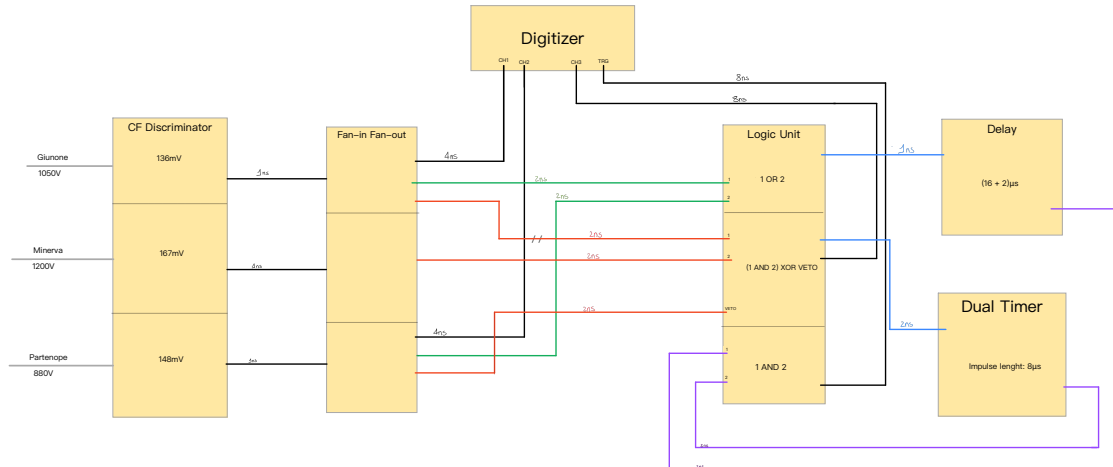
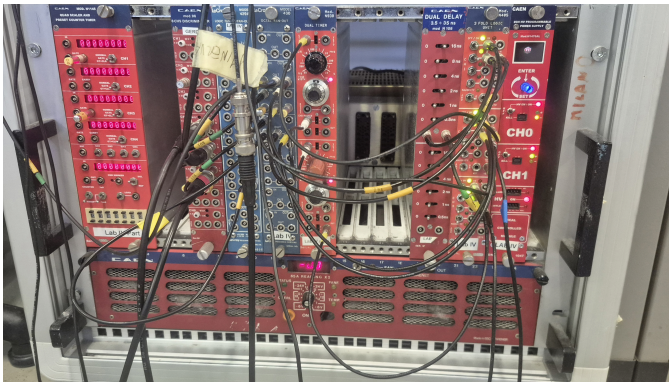
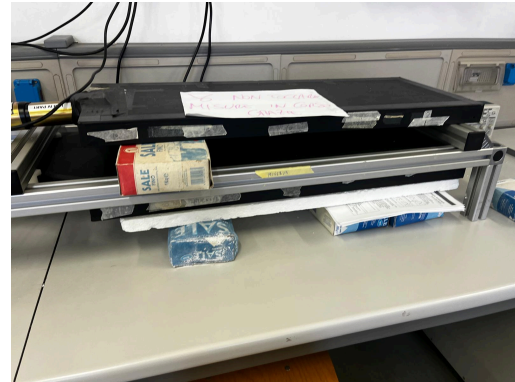


Figure 13: electronic setup graph



(a) circuit



(b) scintillator setup

Figure 14: our setup

This is the result of multiple iterations of possible circuits. The final result can be divided into 3 components:

- Discriminator and Fan-In/Fan-Out
- Logic Unit, Delay unit and Dual Timer
- Digitizer

Discriminator and Fan-In/Fan-Out make out the initial part of the circuit. Their role is to take the analogical input from the PMTs, transform them into logical signals, duplicate them and then send them to the next steps.

The second section of the circuit has the role to compute the logic and manage the delays between logical inputs. The Logical unit is divided in 3 identical sections. The first one detects whether a particle has passed through one of the external scintillators and not in the other. It detects the electron after the decay of the muon. The second one detects if a particle passed through the top and middle scintillator and not in the bottom one. It detects a muon that stops in the middle scintillator. The third one combines the other two into a single one and orders them temporally: it doesn't accept an electron if a muon hasn't happened before in a time range dictated by the dual timer width. In this section, the delay module has the utility of cancelling the delay caused by the dual timer initial stage and by fine tuning these two different windows we are able to not include the muon in the final results since if a signal returns true in the second section, then it will return true also in the first one.

Lastly, the digitizer has the role to transform the logical (or analogical) inputs into digital ones. It uses a trigger to start the data taking window. The window length is uniquely dictated by the buffer length since the frequency cannot be changed. Moreover the digitizer has the possibility to set the trigger to be in a specific part of the buffer: if we choose it to be at the end of the buffer, we will have it as last input of the trace.

This is exactly what we did: we used the third section of the logic unit (the one that tells us if an electron has been detected) as trigger and set the trigger position at the end of the buffer such that we were able to go "back in time" to look for the muon stopping inside the middle scintillator.

This has been the most efficient way to take data by optimizing the void trace percentage.

Since we were interested in measuring the asymmetry between the top and bottom scintillator, we also fed the digitizer the two outputs of the upper and lower scintillators such that we are able to distinguish where the electron landed.

3.1 Minerva pulse issues

We soon noticed that Minerva was the worst of the three scintillator. This was due to its thickness being half of the other two but also due to a phenomena for which pulses coming from Minerva were correlated to spurious correlations in the range of $1\mu s$ with a gaussian distribution.

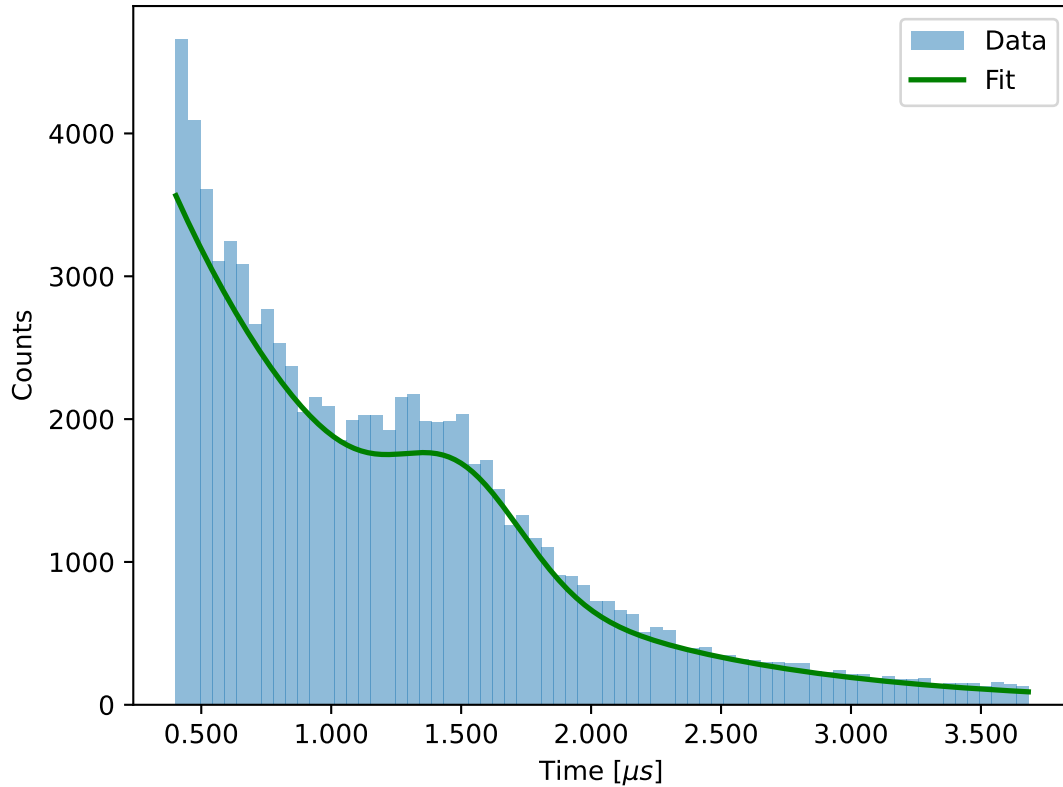


Figure 15: Muon distribution with timestamp taken from only Minerva

This is most likely due to a malfunction in the PMT that at the firsts multiplication stages is instable and therefore lets out electrons that should not be accelerated.

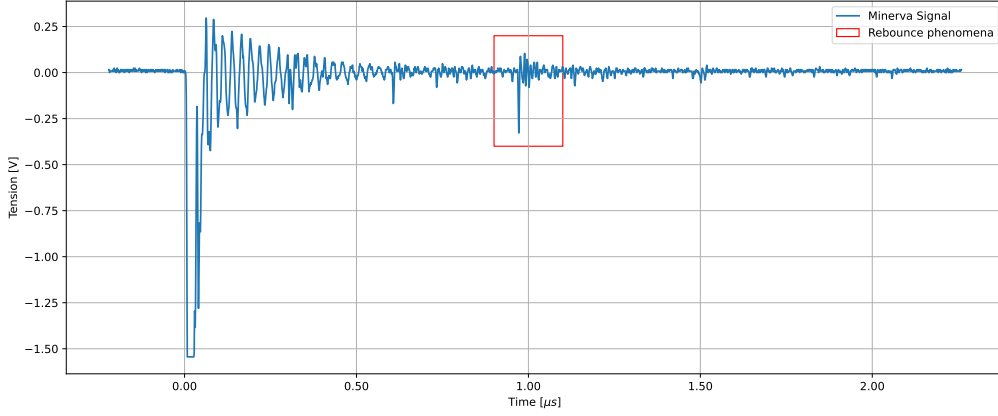


Figure 16: Example of the rebound phenomena in Minerva

We fixed this issue by placing Minerva in between the two others: we willingly choose to have a thinner layer of scintillator material in which the muon can decay with the advantage of having a cleaner signal when it comes to detecting the electron. Since we decided to use the electron as trigger, this resulted also in a decrease of the rates of empty traces

3.2 Time resolution and dead time

We chose to measure the time resolution of the setup by simulating an event with a wave generator. Giving to the discriminator a strictly positive square wave with a width comparable to the signal observed and a low frequency to avoid pileup effect, we were able to delay the signal of the bottom scintillator to simulate both the muon decay and the electron release with only one impulse from the wave generator.

It is enough to simulate the electron energy release only in the bottom scintillator since the electronic chain is completely symmetrical.

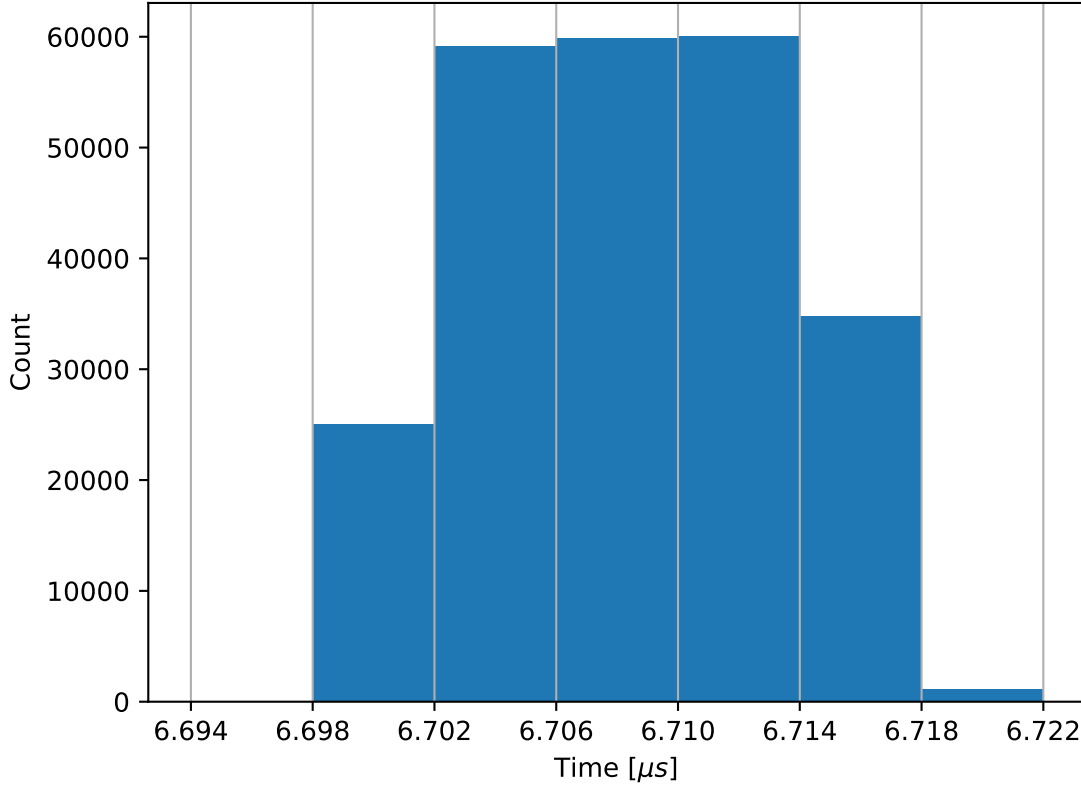


Figure 17: Measured time resolution of the whole setup

The result is a distribution about uniform of width $12ns \pm 4ns$, the uncertainty is fully determined by the frequency of the digitizer. This tells us the minimum error that we can expect from our setup without considering the scintillators.

Another statistically relevant characteristics of the setup is the dead time. Since the setup used previously was not useful for an accurate measurement of the dead time, we opted for calculating this factor using actual data. This is easily done using actual data: if we express the impulse given by the muon stopping in the middle scintillator and we look for the closest electron signal, what we find is the exact measurement of the dead time of the setup

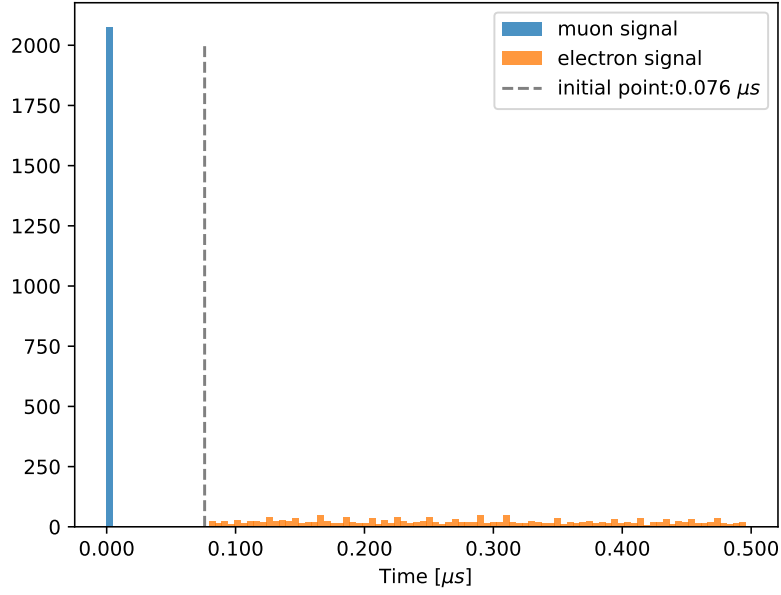


Figure 18: zoom on an initial part of a dataset

We can easily measure it given a dataset and it turns out to be: $76ns \pm 4ns$ where the uncertainty of this value is again dominated by the frequency of the digitizer.

3.3 Pileup probability

Given the dead time calculated previously and the fact that the apparatus is non paralyzable we are able to calculate the expected rate of our setup:

$$n = \frac{m}{1 + m\tau} \quad (3.1)$$

where m is the rate at which we measure events and τ is the dead time of our apparatus. Across all measurements done, we measured an average rate of: $m = 0.012 \frac{\text{muons}}{s}$ and a dead time of: $\tau = 76ns$.

We see that the expected rate n is equal to our actual measured rate and therefore we have a pileup probability of 0%.

4 Scintillators

4.1 Technical characteristics

This chapter details the physical and operational properties of the plastic scintillators used. These detectors, based on a Polyvinyltoluene (PVT) polymer base, are known for their high speed, excellent light yield, and versatility in high-energy physics applications.

The scintillators used (specifically the EJ-200 / BC-408 and EJ-204 / BC-404 series) are organic plastic scintillators. Their operation is based on the excitation of the polymer base by ionizing radiation, followed by the emission of fluorescence photons. Polyvinyltoluene provide high light conversion efficiency. The primary advantage of plastic scintillators is their fast response time, making them suitable for high-count rate experiments:

- Rise Time (t_r): Extremely short, typically between 0.5 ns and 0.9 ns.
- Decay Time (τ): Ranges from 1.8 ns to 2.1 ns (for general-purpose models like BC-408/EJ-200).
- Pulse Width (FWHM): Approximately 2.2 ns to 2.5 ns.

To ensure maximum light collection and protect the detector from environmental interference, the scintillators undergo a specific wrapping process. The plastic material is first wrapped in aluminum foil that acts as a specular reflector. Then a layer of black vinyl or black tape is applied over the aluminum to provide a light-tight seal. This prevents ambient light from entering the system, which would increase the background noise or saturate the PMT.

The thinner scintillator (Minerva) is coupled to the PMT using a BC-802 Tapered Light Guide. Its geometry allows for the transition from the rectangular shape of scintillators to the circular window of the PMT. Minerva has the Hamamatsu R1924A-01 as PMT, a 1-inch diameter tube designed for fast timing and prepared for positive high voltage (HV).

The other two scintillators (Partenope and Giunone) have similar characteristics for which we know only the constructor name (Saint-Gobain), but not the exact specifics. We know they require negative supply voltage.

4.2 Supply voltages & Thresholds

We were now able to test the three scintillators we had: Partenope, Minerva, Giunone. The first interesting thing to study was the relation between number of events, supply voltage and threshold of reading electronics. For a good scintillator-PMT apparatus we expected the output signal to be proportional to the energy released in the interaction by a particle. Thus, at fixed supply voltage, we would expect the higher the threshold is after the PMT, the fewer events measured by a scintillator. For low threshold we expected many counts due to natural radioactivity, but we were interested in measuring the decay of the muon. Its energy scale is one order of magnitude greater than the one of natural radioactivity. Therefore, we expected some kind of plateaux where the natural radioactivity ends and there is only a constant rate of muons.

We first made some preliminary measurements. These measures were taken varying both the supply voltage and the threshold of the discriminator at the same time (or only one of them) for the three scintillators. These data helped us understanding the number of events we would expect and how much it should have been varying to different initial condition. The first idea was to map the two dimensional space of tension and threshold at the same time, but we soon realized it was a waste of energies and time. Thus, we decided to test the distribution of events at fixed supply voltage, increasing the threshold. We chose the supply voltages by visually analyzing the output of PMTs at the oscilloscope and maximizing it with respect to the maximum input the discriminator accepted (250 mV). Then, eventually, we would have lowered the supply tension and retake the measures. In this way, we were sure to respect the requirements of the instrumentation and be able to test all the scale of threshold our system had, searching for the plateaux. This was done for each scintillator.

For what it concerns Giunone and Partenope scintillators, from Figure 19 it is evident that in this setup there were no plateaux. We made some hypotheses: it could be that we missed the plateaux because we took too big steps in threshold resolution or that it was at lower or higher threshold we couldn't set or it could be that the supply voltage wasn't high enough.

We also studied the coincidences between the two, trying to maximize the ratio of coincidences and total number of events. But after some measurements we realized this

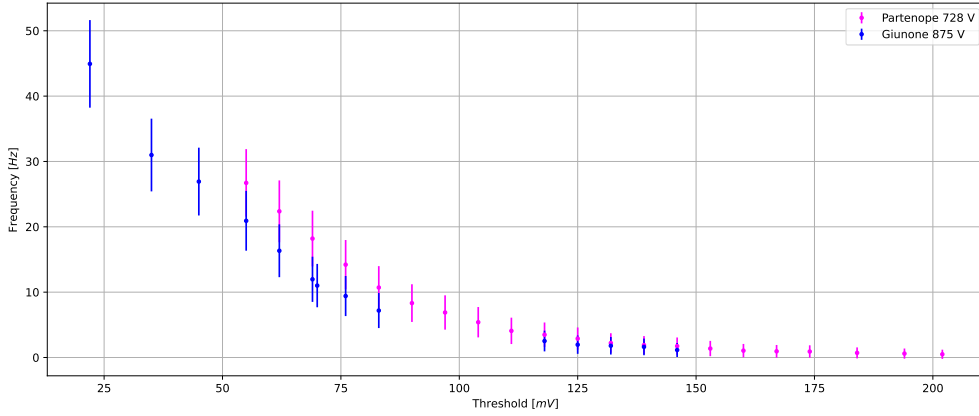


Figure 19: Partenope and Giunone's count vs threshold

procedure was too messy and wasteful because it depends on the thresholds and supply voltage of both scintillators. Plus, it also could lead us to some misinterpretation of the data because we only measured the double coincidences, so they were affected by casual coincidences and natural radioactivity (since we did not know the number of events from muons' interactions). After having denied the idea of having missed the plateaux because of far too big steps, we thought that maybe the supply voltages of our instruments weren't high enough, so PMTs couldn't pre-amplify correctly the signal, which could lead to inaccurate signals. Thus, as discussed in Sec. 2.5, we changed the discriminator in order to widen the operational range.

We were now able to scan the number of events respect to the thresholds at two different voltages. After analyzing the output of the detectors at the oscilloscope, we noticed that in order to have similar responses to the same input, Giunone detector had to be powered with ~ 100 V more than Partenope. We decided to have similar impulses to the same particle interaction.

As it can be seen from the plots (Figure 21 and Figure 20), for each of the two scintillators, we scanned the threshold at two different voltages: for the higher ones tested, we were able to recognize a plateaux for threshold between 130 mV to 175 mV. The plateaux isn't perfectly flat, but we can expect to cut some events going at higher thresholds (due to the energy distribution or fluctuations). Anyway, we decided to set the threshold at half of the plateaux.

Minerva detector was already at the max supply voltage we could set (for PMT requirements), so we didn't change its one. As can be seen from the plot (Figure 22),

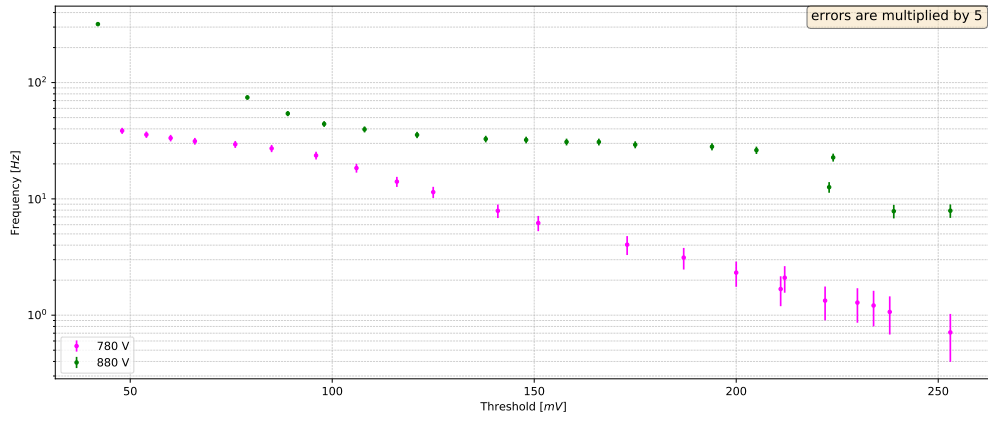


Figure 20: Partenope's threshold

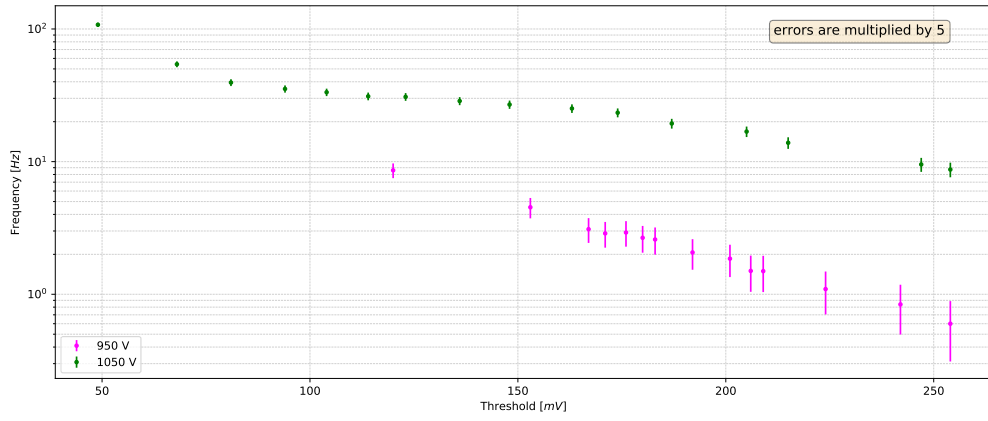


Figure 21: Giunone's threshold

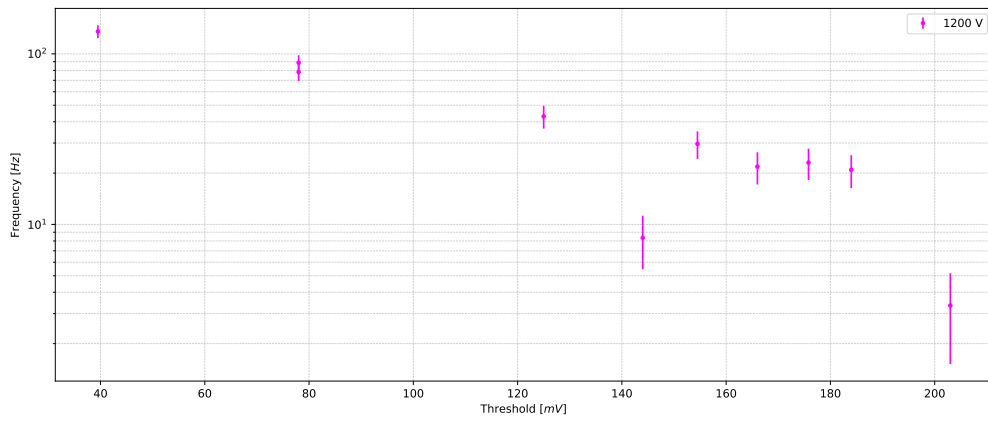


Figure 22: Minerva's count vs threshold

for this scintillator we found some kind of plateaux. The plateaux was observed for Minerva scintillator from 157 to 186 mV.

	Supply voltage [V]	Threshold [mV]
Partenope	880	148
Giunone	1050	136
Minerva	1200	167

Table 5: Scintillators operating characteristics

4.3 Efficiency

Having decided the supply voltages and thresholds of the setup, we then decided to study the efficiencies of our instruments: we wanted to test our scintillators to understand the fraction of event registered respect to the number of total interactions. We did this for different configurations to verify also that no scintillator had any “dead zones”. We couldn’t know the absolute number of interactions, but we estimate it by using the counts of coincidences of the upper and lower scintillators. We defined

$$\varepsilon = \frac{\# \text{ triple coincidences}}{\# \text{ upper and lower scintillators coincidences}} \quad (4.1)$$

We put in the middle the scintillator we wanted to test and above and below the others. For a fixed time, we measured the number of triple coincidences of the three instruments and the number of double coincidences between the upper scintillator and the lower one. We then moved at fixed steps the central scintillator and retake the measurement. We ran a simulation to understand the number of data lost for solid angle (considering different distances among them) and we compared the results. Eventually, we repeated the procedure with the other two scintillators.

4.3.1 Simulations

We ran two simulations: the first was a Python simulation in which we considered only the solid angle factor, the other was a Geant4 simulation in which we simulate real interactions. For the first one, we wanted to have an order of magnitude of muons that interact with the upper and lower scintillators while misaligning the three instruments. With this, we did not considered a minimum path and energy released in each of them, but simply if a muon crossed the surface of the scintillator. The second one was a more complete simulation, where we could set energies of particles, kind of interactions, minimum energy released. The two simulation aren’t fully compatible, but we do trust

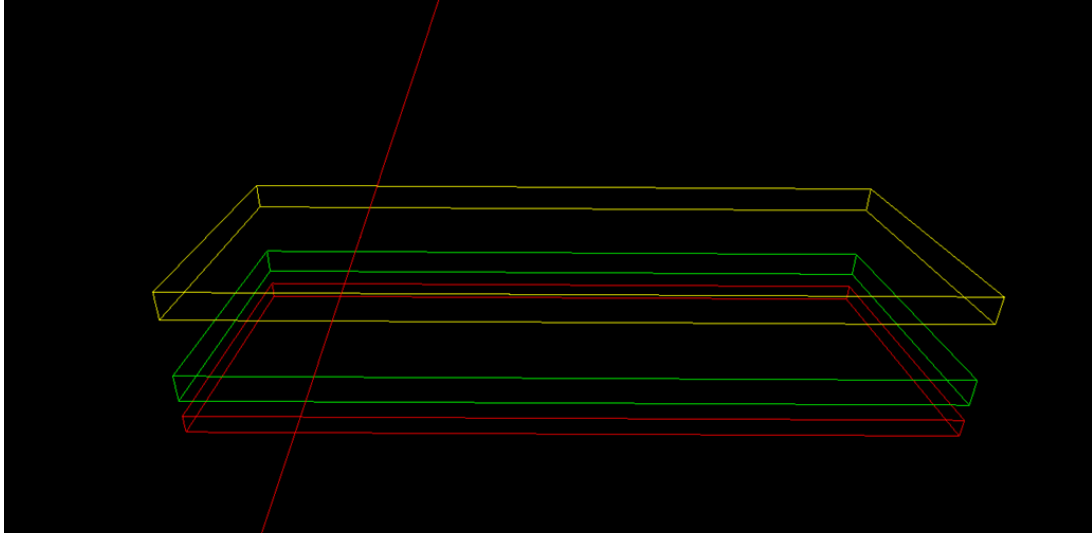


Figure 23: Example of Geant4 simulation run: Minerva (yellow), Giunone (green), Partenope (red)

more the Geant4 simulation because of its reliability and completeness. With this last simulation we also set different muons distributions (as described in Sec. 1.3) to understand which one cosmic muons have.

4.3.2 Partenope & Giunone

For those scintillators we did two different simulation in which we varied the length of this detector: in the first simulation it was 80 cm, in the second one it was 75 cm. Since we do not have any technical scheme of the scintillator, we tried to understand how much 5 cm of light collector near the PMT would affect the measure. As can be seen in the following plots (Figure 24 and Figure 25) the difference is very small, almost imperceptible. In these simulations we tried also different angular distribution of cosmic muons respect to the azimuthal angle, but none of them is really compatible with what we found experimentally.

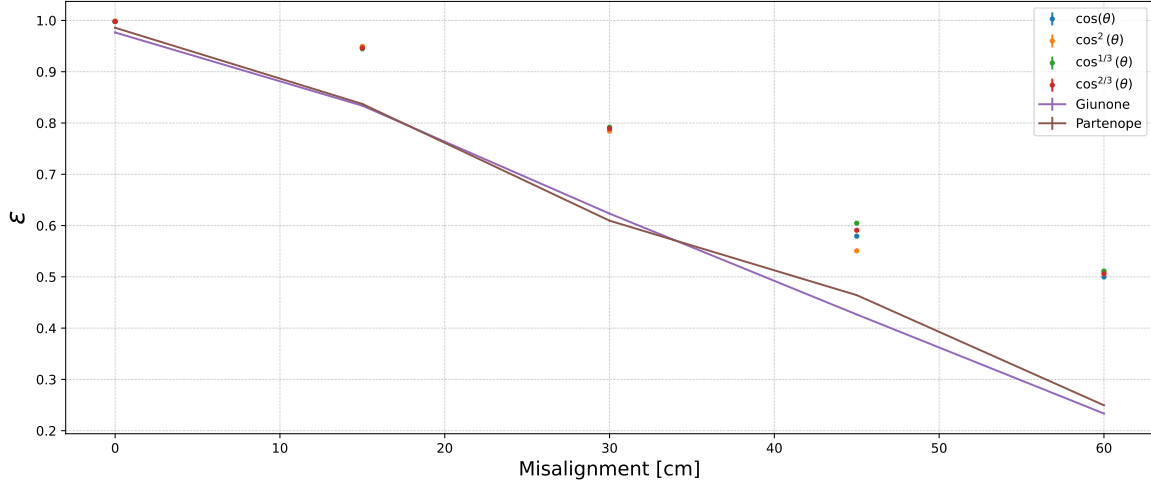


Figure 24: Partenope and Giunone resolutions, 80 cm length

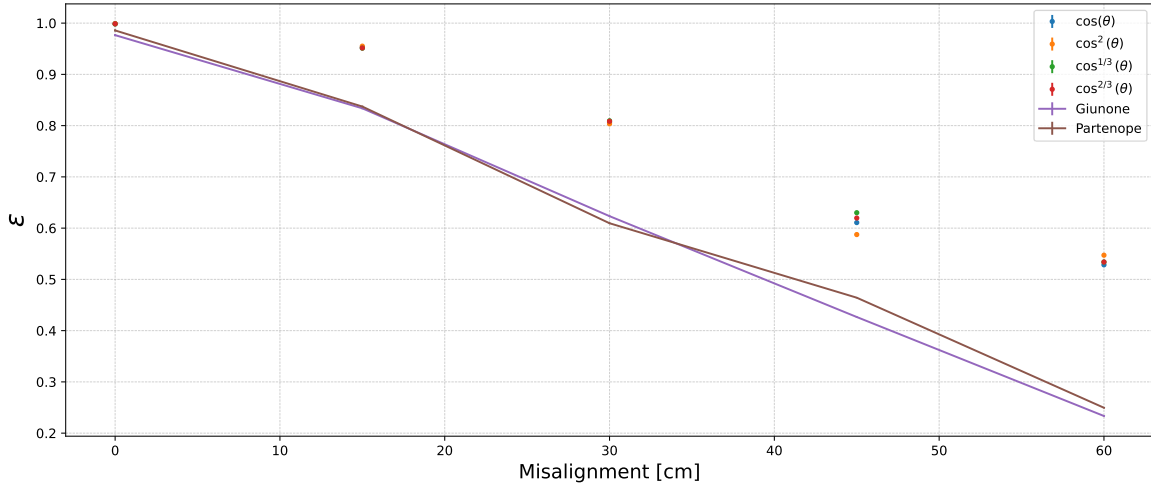


Figure 25: Partenope and Giunone resolutions, 75 cm length

If we define the geometrical efficiency as the fraction $\frac{\varepsilon_{\text{measured}}}{\varepsilon_{\text{theory}}}$, with ε defined in Eq. (4.1), both Partenope and Giunone's plots have an unexpected shape (see Figure 26, Figure 27, Figure 28, Figure 29). The efficiency drops almost linearly with the misalignment of the middle scintillator, even if it is about 98% (Giunone) and 99% (Partenope) when aligned. This makes us think that there can be some dead zones in this two scintillators, but we can't locate them inside the detector. This non-scintillating parts of the detector can be both non-functioning PVT volumes or non-scintillating components of the apparatus PMT-scintillator, like light guides or wrapping.

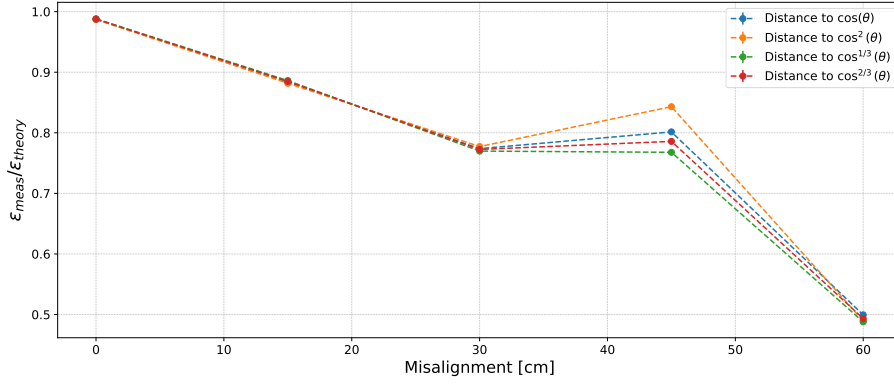


Figure 26: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Partenope, 80 cm length

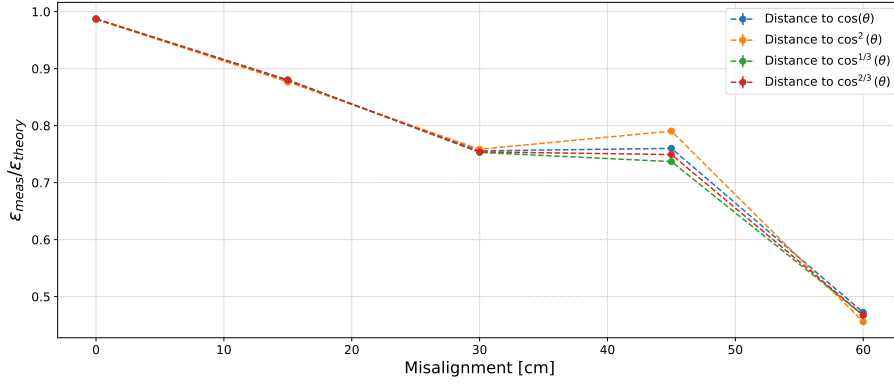


Figure 27: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Partenope, 75 cm length

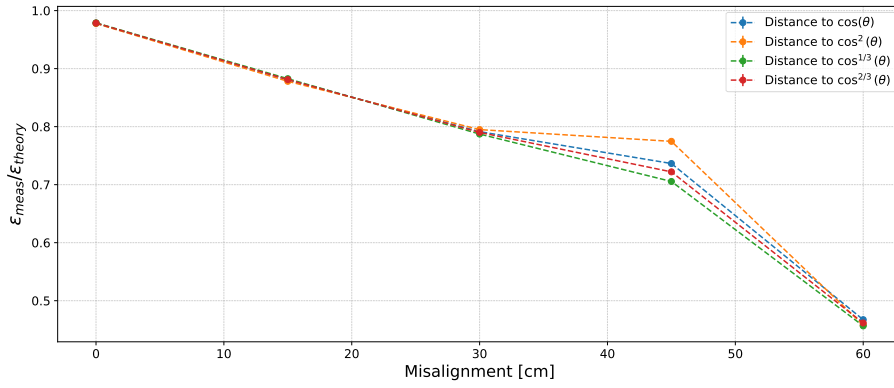


Figure 28: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Giunone, 80 cm length

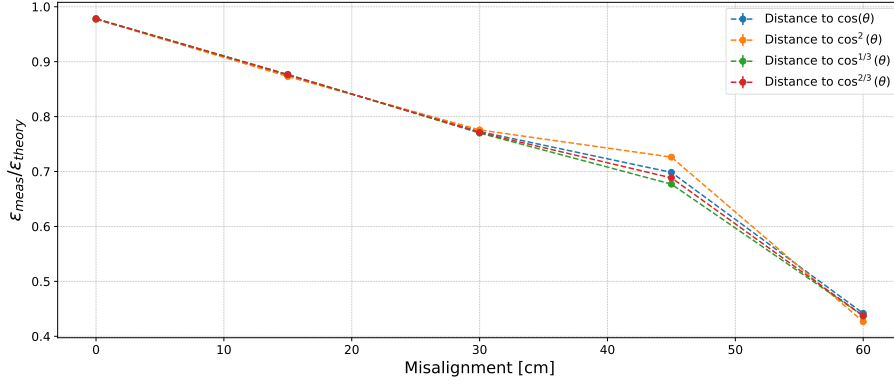


Figure 29: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Giunone, 75 cm length

Even though there are dead zones in Partenope and Giunone, we are satisfied by the high efficiency when they are aligned because this is the setup for then next measures.

4.3.3 Minerva

We can observe the plots for Minerva scintillator (Figure 30 and Figure 31): this detector had a resolution of about 89% when completely aligned with the others. The “75 cm” label indicates that the length of Partenope and Giunone is 75 cm for that simulation (generated with a cosine distribution) because of what we said in the previous paragraph. The resolution remains constant or slightly increase until 45 cm of misalignment and then it drops at about 60-70%.

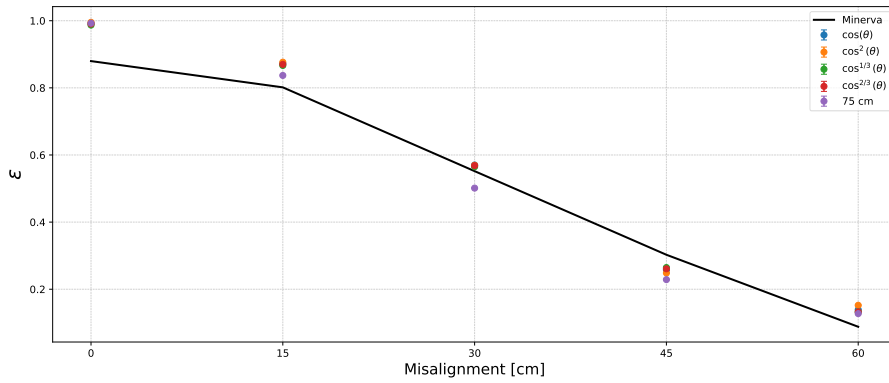


Figure 30: Minerva resolution

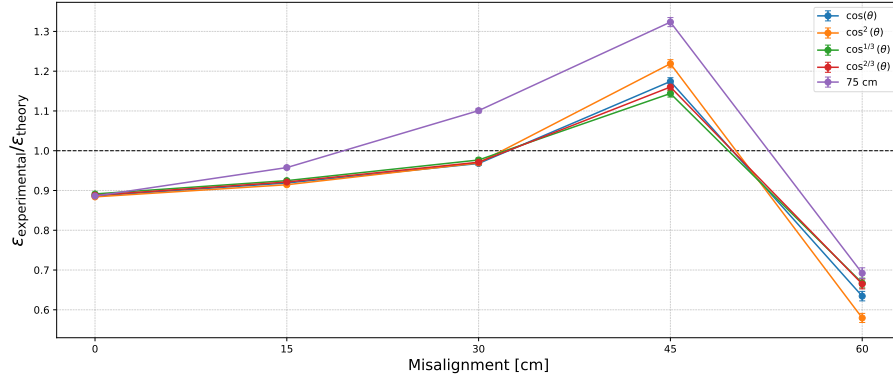


Figure 31: $\varepsilon_{\text{meas}} / \varepsilon_{\text{theory}}$ for Minerva

There is a sort of bump in this last plot where the efficiency is greater than 1, which means we measured an higher count of triple coincidences than the one we would expect due to solid angles. We can see a similar pattern also in the same plots for the other two scintillator. We can address this to the problems we saw with Minerva's PMT or it might be some interference of the PMT while it is in the middle of the other scintillators or the theoretical distributions don't describe our setup correctly. Also here we see that the dimensions of the scintillators are important, so ignoring the real dimension of two scintillators doesn't help understanding how performative Minerva is.

5 Data Analysis

The data analysis has been made using iMinuit, we used poissonian errors for each bin and we minimised the negative log likelihood. All analysis and more is available on github [1]

5.1 Initial issues

We found three issues with this method:

1. what to use to quantify the goodness of the fit
2. how many bins would each histogram need
3. the model to choose

5.1.1 Chi square as estimator of the goodness of the fit

Although data inside each bins are distributed as a poissonian distribution, with enough data we can fall back under the hypothesis of data distributed as a normal distribution. If we assume this hypothesis as valid, what we found is a minimum number of events per bin to approximate the poissonian distribution to a normal distribution and, if we respect this constraint in each bin, we can say that $\tilde{\chi}^2$ is a good estimator of the goodness of the fit, similarly to what happens when using Least Squares method. Therefore we decided to use $\tilde{\chi}^2$ as an estimator of the goodness of the fit.

5.1.2 Model of the interpolation

We were uncertain about whether to consider an uniform noise background or to interpolate without it. By letting the frequency between exponential distributed points and uniform distributed points, we found that the noise frequency is around 5%. Although it might be small, it has a relevant impact in the error estimation of the variable: if we consider an uniform background, both relative errors and $\tilde{\chi}^2$ decrease, therefore we decided to keep it.

5.1.3 Bins choice

To choose how many bins to give each histogram, we used $\tilde{\chi}^2$ to see how the interpolation behaved at different bins width. Therefore we chose the number of bins that minimises

the $\tilde{\chi}^2$. We found it generally to be between 60 and 100 bins but the specific number changes from dataset to dataset.

5.2 Vacuum measurements

As said in Sec. 1.3, if the muons decay in the scintillator, a very light difference in the time characteristic is present. This difference cannot be seen in our data sample since it is dominated by statistical uncertainties. We therefore compare the result of the fit to the decay time of the muon in vacuum.

As fit function we use a combination of exponential and a uniform background, normalised on the dataset, as follows:

$$f_{\{\text{vacuum}\}}(t; f_{\text{exp}}, \tau) = N \left[f_{\text{exp}} \cdot p_{\text{exp}}(t, \tau) + (1 - f_{\text{exp}}) p_{\text{uniform}} \right] \quad (5.1)$$

where p_{exp} and p_{uniform} are the CDF of exponential and uniform distributions limited to the range of the dataset and N is the total number of events.

Due to the presence of unexpected signals at the start of the trace, and due to edge effect at the end of the dataset, we decided to apply two cuts to the dataset. To choose the optimal position where those cuts have to be performed, we ran some interpolations varying the time interval in which we consider the data. From every fit, we took the $\tilde{\chi}^2$ and showed it in Figure 32.

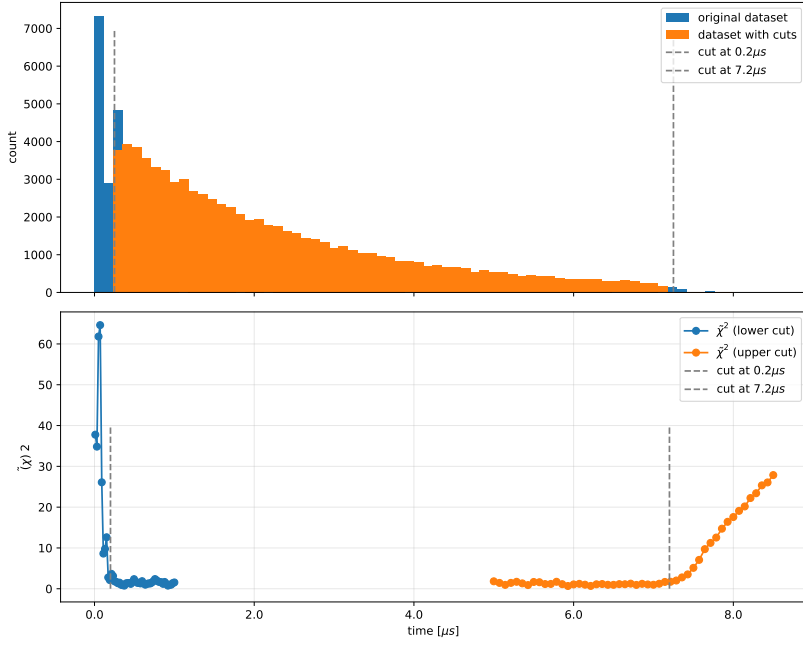


Figure 32: Vacuum analysis:

On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the edges of the histogram

We can see how at the start and at the end of the dataset there are data that notably worsen the interpolation. We believe the signals in the very first $0.2\mu s$ are due mainly to Minerva's PMT fluctuations overlaid on muon's decay signals. The noise over $7.5\mu s$ is due to a small jitter in the end gate of the timer that therefore decrease the number of data acquired in that region and therefore decide to remove them.

In order to choose the number of bins N to use, we calculated the $\tilde{\chi}^2$ of the dataset for different number of bins and chose the one that minimises it.

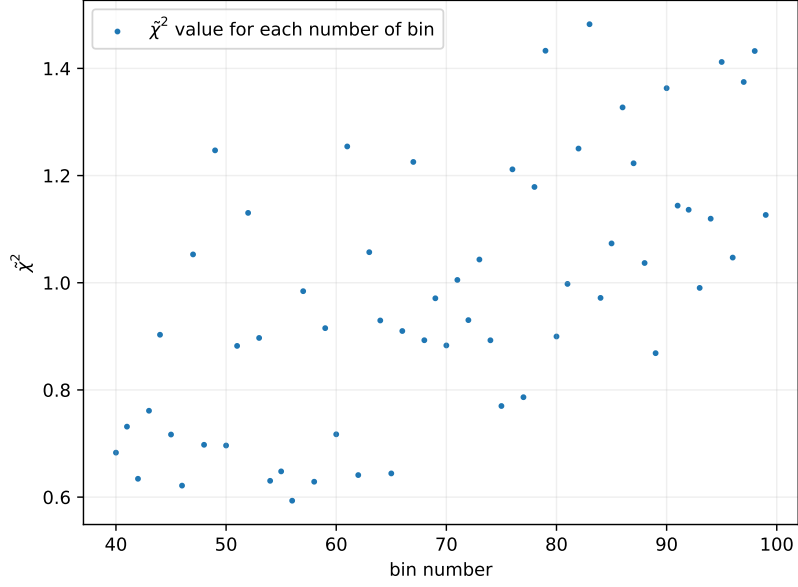


Figure 33: Vacuum analysis: $\tilde{\chi}^2$ for different number of bins.

Since we don't see a clear pattern varying the number of bins, we decided to take 68 bins, for the range $[0.2, 7.5]\mu s$, causing every bin to have a $10ns$ width. In Figure 34 we show the final fit of the dataset, together with the percental deviation of the data from the interpolated function. Table 6 contains the fitted parameters of the free decay, with their errors.

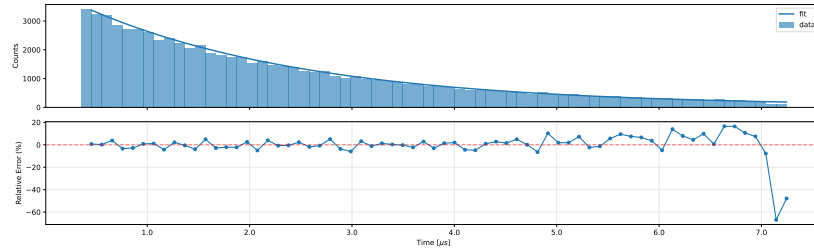


Figure 34: Vacuum analysis:

On top: The fitted dataset, and the resulting interpolation function.

On bottom: Relative deviation of the data respect to the fit.

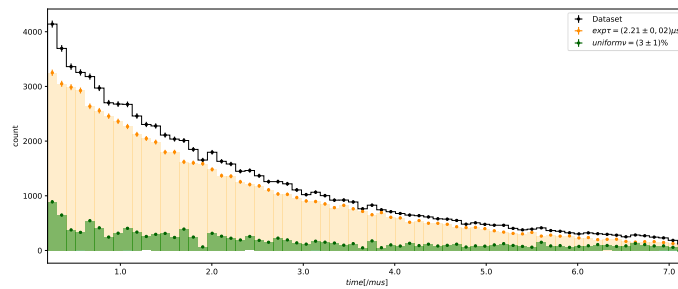


Figure 35: Vacuum analysis: Simulated contributions of the vacuum dataset and real dataset

In the figure above we show how the sample are distributed according to the fitted parameters of the functions. The components of the plot are simulated but it gives a good idea of how frequent the data were part of each phenomena.

Parameter	Value	Uncertainty
f_{void}	0.97	0.010
τ_{void}	$2.193\mu s$	$11ns$

Table 6: Vacuum fit parameters

From the values in Table 6, and using the function in Eq. (5.1), we can derive the amplitude A of the uniform background, and its error, obtaining $A = 1 - f_{\text{exp}} = 0.033 \pm 0.010$. Although the uniform background noise is low, it is not compatible with zero, therefore is not negligible.

In Figure 36 we also show the Negative-Log-likelihood of the parameter τ , and the comparison with the expected value.

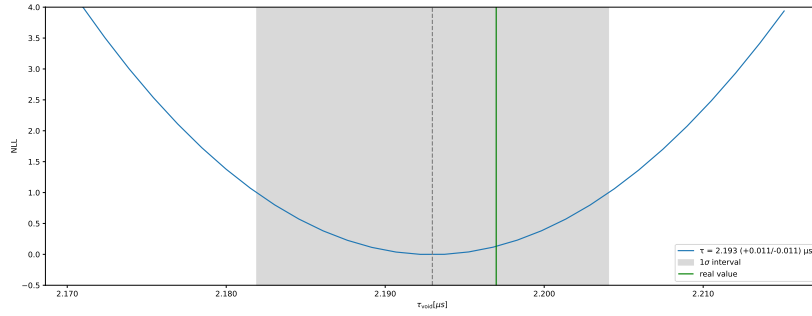


Figure 36: Vacuum analysis: Negative-LogLikelihood of the parameter τ_{void}

As we can see, the expected value is below 1σ distant from our measure, therefore is compatible.

As a further check for our setup, we looked at the up-down asymmetry of this dataset, defined as in Eq. (1.12).

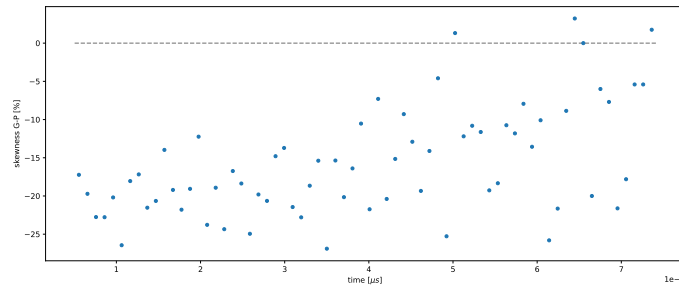


Figure 37: Vacuum analysis: Up-Down asymmetry

We can see in Figure 37 how a relevant asymmetry in favour of the lower scintillator is present. This type of asymmetry will be present across all measurements and is most likely due to a miss-calibration of the scintillators.

5.3 Decay time in salt

We changed our setup to accommodate salt in between Minerva and Partenope. This has brought a different behaviour to be studied: the decay of muons in salt due to the interaction with the nuclei. This interaction is way more frequent than in carbon and it creates a greater variation with respect to the mean lifetime of a muon (Sec. 1.5).

To study this behaviour, we modified the model used previously in Eq. (5.1). In particular, we switched the exponential CDF with a double exponential determined by another two parameters of the interpolation

$$f_{\text{exp}} \cdot p_{\text{exp}} = f_{\text{void}} p_{\text{exp}}(t, \tau_{\text{void}}) + f_{\text{NaCl}} p_{\text{exp}}(t, \tau_{\text{NaCl}}) \quad (5.2)$$

where p_{exp} is the CDF of exponential distribution limited to the range of the dataset and the factor in front of the uniform CDF has transformed accordingly.

Similarly to what we did with vacuum measurements, we used $\tilde{\chi}^2$ to decide bin width and cuts to apply at the dataset. Since the setup is the same as before, here we expect to apply cuts in points similar to the ones in Figure 32.

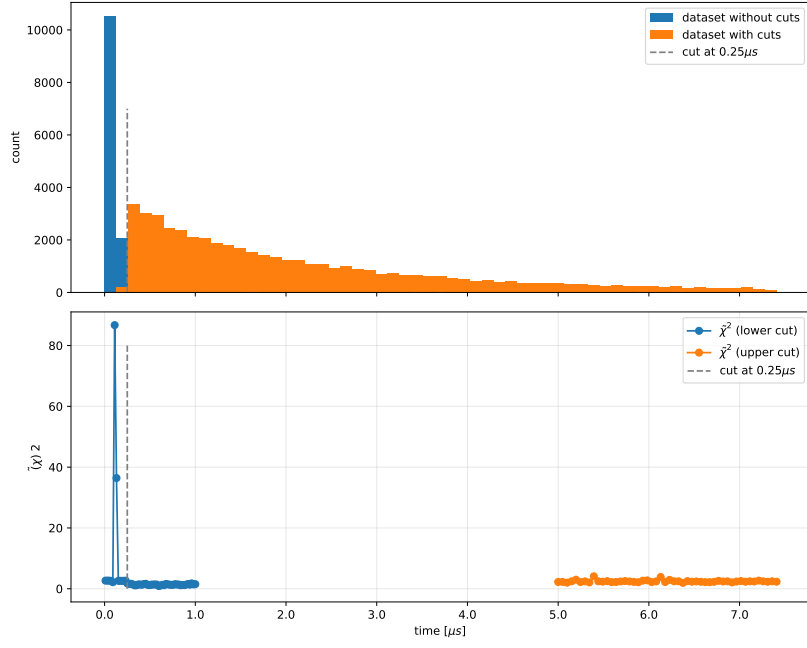


Figure 38: salt analysis:

On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the edges of the histogram

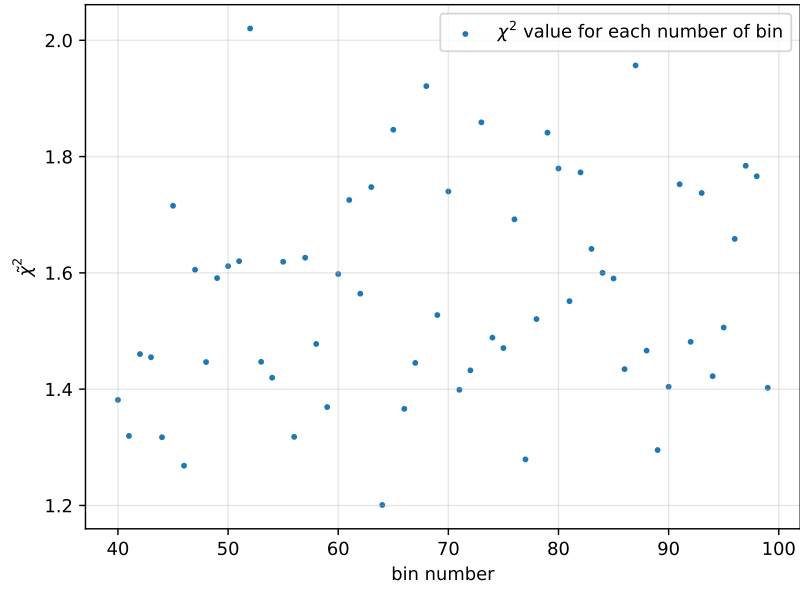


Figure 39: salt analysis: $\tilde{\chi}^2$ for different number of bins.

Following the same logic as the previous case, we opted for a bin width slightly larger with a number of bins of 64 and a bin width of $11.4ns$ in the range $[0.2, 8]\mu s$.

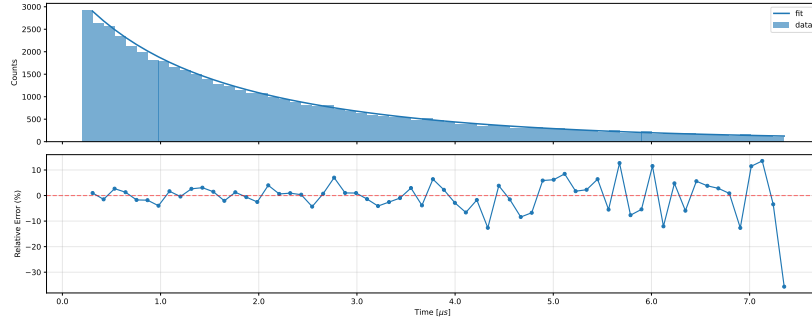


Figure 40: Salt analysis:

On top: The fitted dataset, and the resulting interpolation function .

On bottom: Relative deviation of the data respect to the fit.

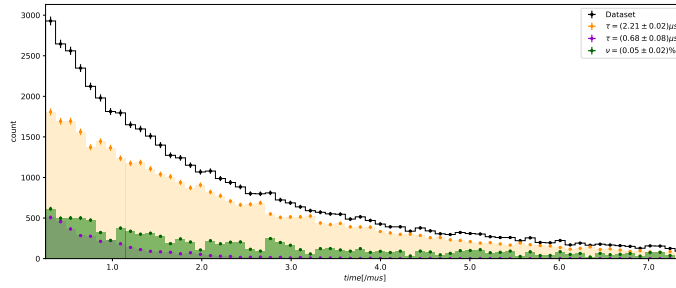


Figure 41: Salt dataset: Simulated contributions of the salt dataset and real dataset

As we can see, here clearly a second exponential is present in addition to the noise background that we expect to be about constant in all the measures taken.

Parameter	Value	Uncertainty
f_{void}	0.8452	0.0217
f_{NaCl}	0.1016	0.0156
τ_{void}	$2.193\mu s$	$22.0ns$
τ_{NaCl}	$700ns$	$75ns$

Table 7: Salt fit parameters

In the interpolation we opted for fixing the components in common between the two measurements. It being the the characteristic of the exponential for the decay of the muon with no interaction. With this parameter fixed, we measure a mean lifetime of the muon in the salt compatible with the expected value that we assume known without error. And a frequency of the noise background similar to the previous case: $A = 1 - f_{\text{void}} - f_{\text{NaCl}} = 0.06 \pm 0.02$

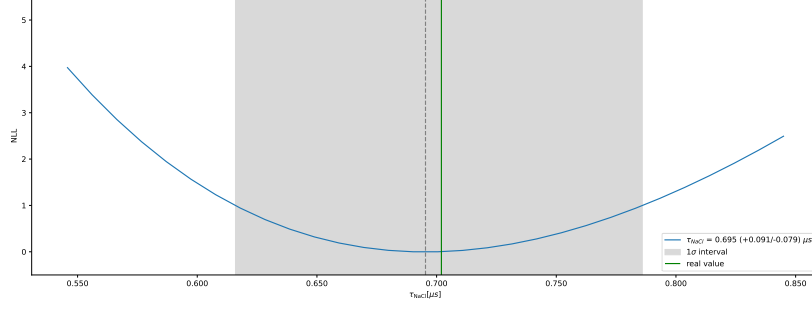


Figure 42: Salt analysis: Negative-LogLikelihood of the parameter τ_{NaCl}

Moreover we kept checking whether the addition of material inside the setup broke the symmetry we found in the previous part.

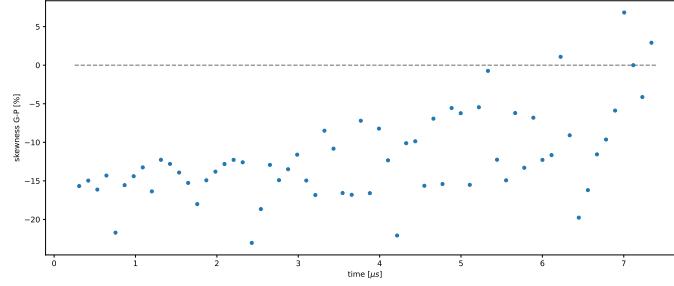


Figure 43: Salt analysis: Up-Down asymmetry

What we found is an asymmetry compatible to the one before.

5.4 Decay time in aluminium

Lastly, we substitute salt with slabs of aluminium. This change, as said in Sec. 1.5, should return data close to the ones gathered in salt but with different frequency and lifetime since aluminium has a higher atomic number.

In this measurements we used the same fit function as in salt, as stated in Eq. (5.2)

Analogously to what has been done previously, we used $\tilde{\chi}^2$ to decide bin number (or bin size) and which cuts to apply.

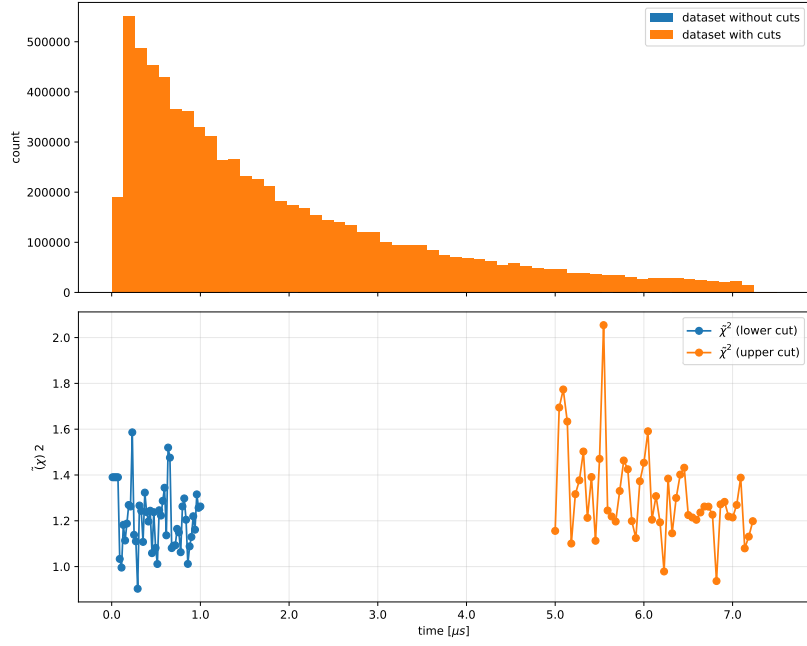


Figure 44: Aluminium analysis:

On top: Raw data collected and the data taken after the cuts

On bottom: $\tilde{\chi}^2$ for different cuts at the edges of the histogram

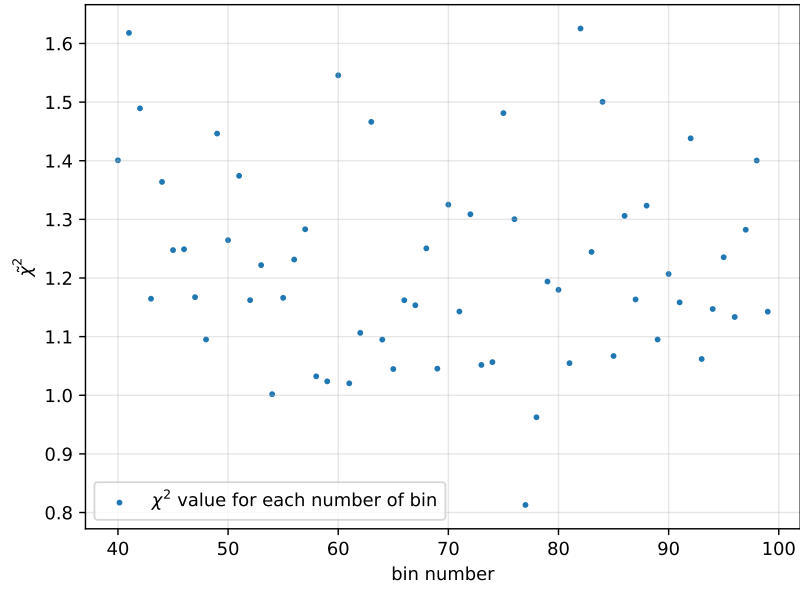


Figure 45: Aluminium analysis: $\tilde{\chi}^2$ for different number of bins

This time, the shape of the dataset has notably changed and with that also the bin number. We chose 78 bins for the range $[0.25, 8]\mu s$ for a bin width of $10ns$

In this case we delayed slightly the electron signal and this gave a huge improvement in the cutting zones. While in the bin width the range is close to the others with 78 bins.

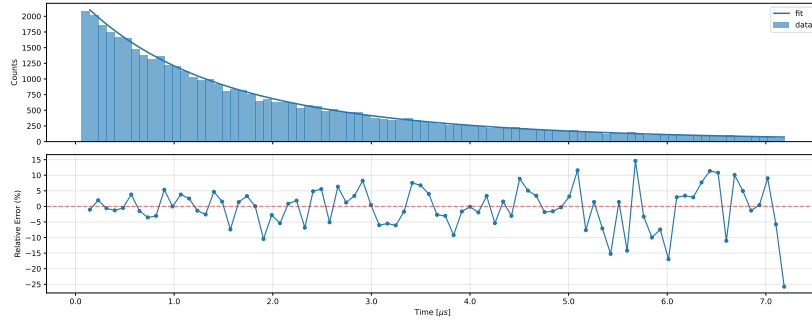


Figure 46: Aluminium analysis:

On top: The fitted dataset, and the resulting interpolation function Eq. (5.2).

On bottom: Relative deviation of the data respect to the fit.

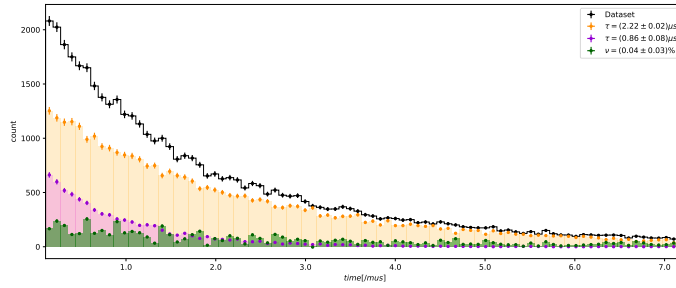


Figure 47: Aluminium analysis: Simulated contributions of the aluminium dataset and real dataset

As previously, a second exponential is present and the uniform distribution is around the same percentage

Parameter	Value	Uncertainty
f_{void}	0.7864	0.0337
f_{Al}	0.1785	0.0262
τ_{void}	$2.193\mu s$	$20ns$
τ_{Al}	$870ns$	$84ns$

Table 8: Fit Parameters

Similarly to what done previously, we fixed the common factors between interpolation. Doing so we found a value of the lifetime of the muon when interact with the nuclei of aluminium that is compatible to what has been found in chapter Sec. 1.5

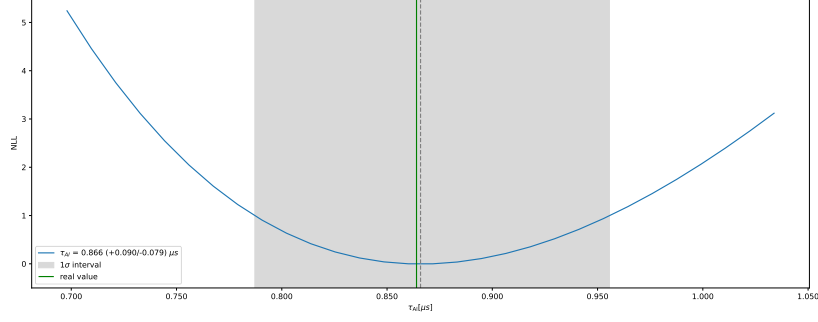


Figure 48: Aluminium analysis: Negative-LogLikelihood of the parameter τ_{Al}

The changes in the setup improve further the symmetry of the setup. In fact, in this dataset we are close to 5% asymmetry of the data taken.

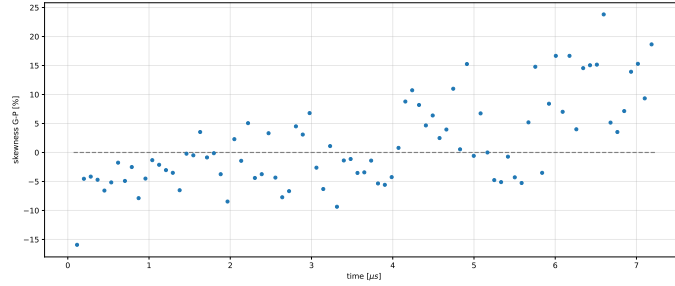


Figure 49: Aluminium dataset: Up-Down asymmetry

5.5 Parity Violation

5.5.1 Setup

As discussed in section Sec. 1.7, in order to see the effects of parity violation with our setup we have to insert the middle scintillator (the one in which the particle decays) in a magnetic field such that the spin of the decaying particle precedes around the direction of the field. Because of mechanical and scheduling issues, we had to join another group that was doing the same experiment and to study parity violation with a different setup: from our previous experiments we only used Partenope scintillator. All the characterisation and tuning of the new electronic setup was done by the other group [14].

We setup the scintillator as done previously in Figure 14b with Partenope in the middle inserted in the magnetic field and the two scintillator of the other group on top and bottom. This choice has been made to be neutral to different calibrations done by the two groups separately.

5.5.2 Calibration of the magnetic field generator

Before starting to take measurements, we need to study the solenoid and be sure that the magnetic field is linear according to Biot-Savar law.

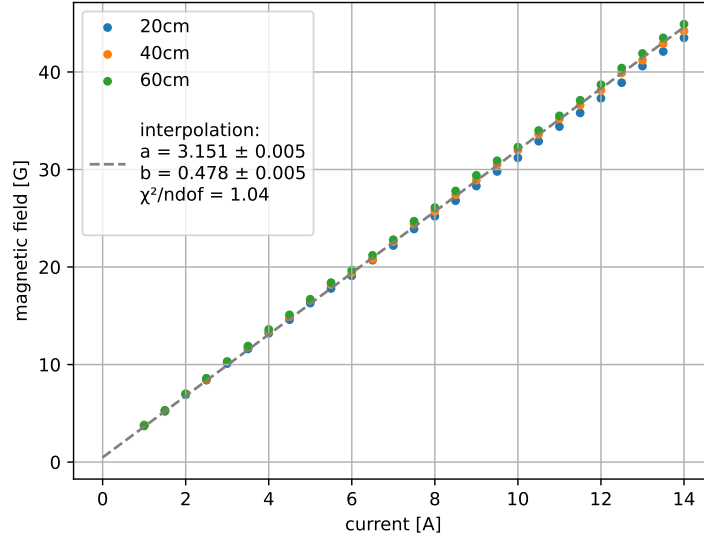


Figure 50: Intensity of field in 3 points of the coil given different currents

Before the beginning of the measurements of the parity violation, we tested the uniformity and the stability of the magnetic field inside the solenoid with a magnetometer. Results are in Figure 51.

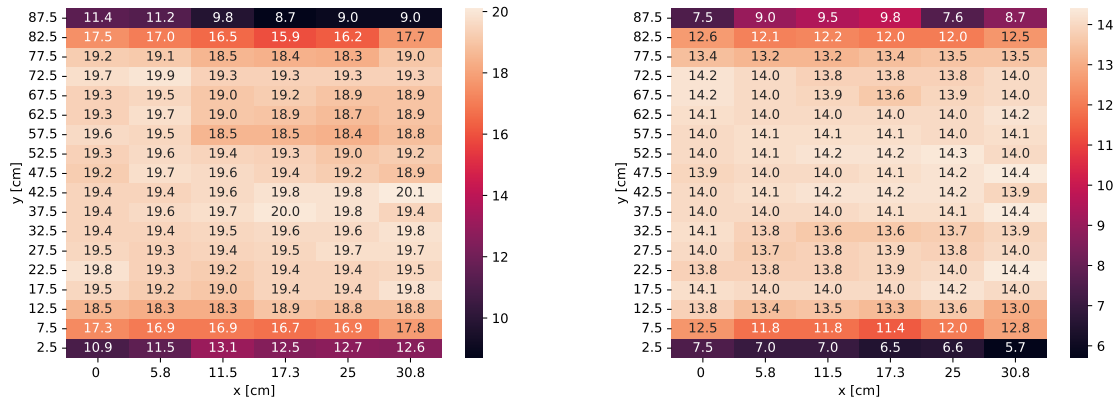


Figure 51: Uniformity test results of magnetic field. The scale is in Gauss. All data are $\pm 0.2G$,

left figure: 6A , right figure 4.5A

As we can see, at high current values, a hole is present in the back of the solenoid. Therefore we limit ourselves to a current of 4.5A that shows no evident sign of non-uniformity.

5.5.3 Measurements with no magnetic field

Since neither of the two group has ever used this setup, we opted for a preliminary measure with the coil but without turning the magnetic field on. This will give us a confirmation of the behaviour of the setup and an initial measurement of the setup asymmetry that will be detracted from the real measurement.

We proceeded therefore in similar way to what has been done previously. Since the objective of the measurement is to check the proper functioning of the system and measure its asymmetry, we will neglect hypothesis testing and accurate measurement of the muon time characteristic.

We therefore proceeded to decide where to cut the dataset and the optimal bin width as before.

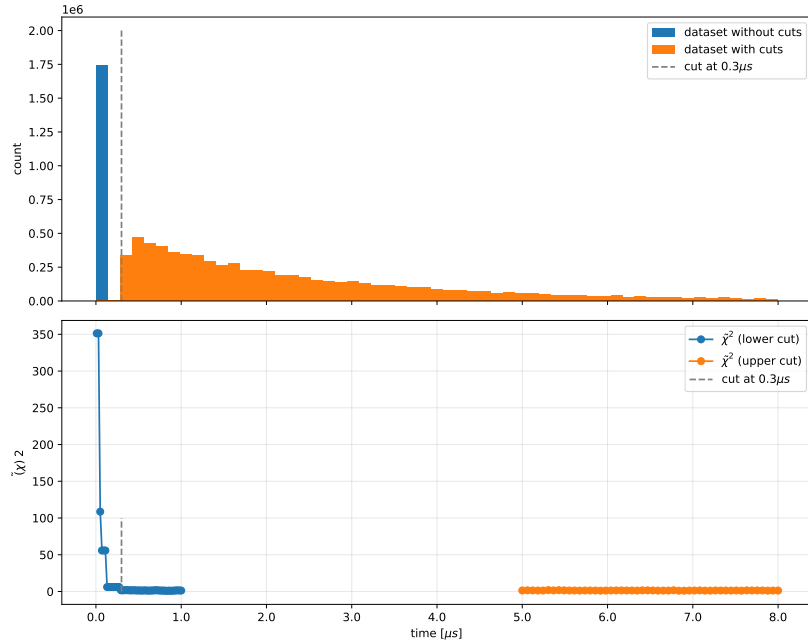


Figure 52: parity violation analysis:

On top: Raw data collected and the data taken after the cuts.

On bottom: $\tilde{\chi}^2$ for different cuts at the edges of the histogram

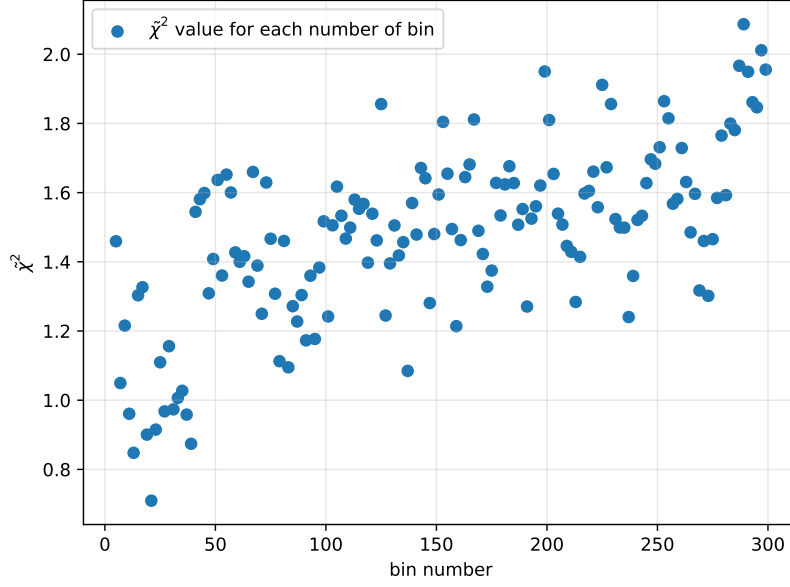


Figure 53: parity violation analysis: $\tilde{\chi}^2$ for different number of bins.

We see that the optimal number of bins is at 80 between the range $[0.3, 8]\mu s$ with a bin width of $10ns$.

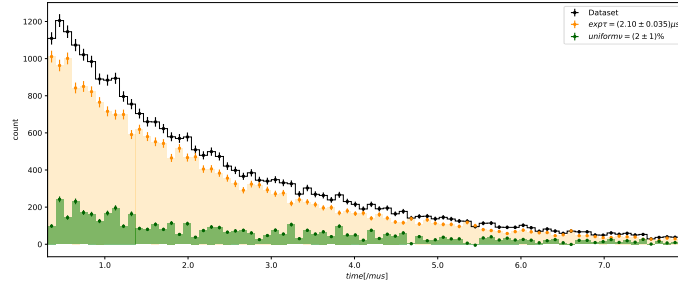


Figure 54: parity violation analysis: Simulated contributions of the asymmetry dataset and real dataset

We see how the uniform is less relevant than the other cases. This gives us reason to think that the uniform noise is mainly given by the presence of Minerva (the thin scintillator), assuming the behaviour of Giunone similar to the one of Partenope.

Parameter	Value	Uncertainty
f_1	0.98	0.01
τ_{void}	$2.095\mu s$	$35ns$

Table 9: calibration fit parameters

In this case we are unable to fix any factors since the setup is different.

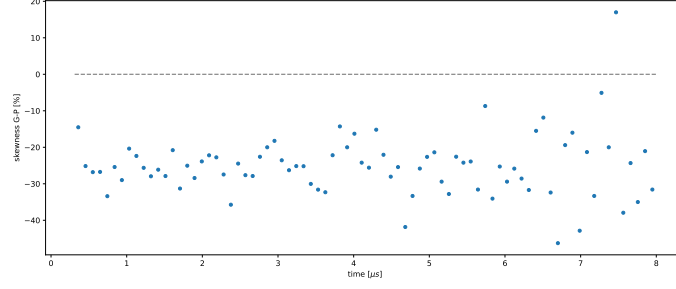


Figure 55: parity violation analysis: Up-Down asymmetry

Finally, we see how in this setup an initial asymmetry towards the bottom scintillator is present and therefore needs to be subtracted from the dataset with the magnetic field on.

5.5.4 Measurement with magnetic field

Turning on the magnetic field, we want to measure the variation of symmetry between the above and below scintillators. If we chose a bin number similar to the one used above, we wouldn't be able to see anything since the difference would be negligible with bins of that size. Therefore we opted for a number of bin of 7 to increase the visibility of the phenomena.

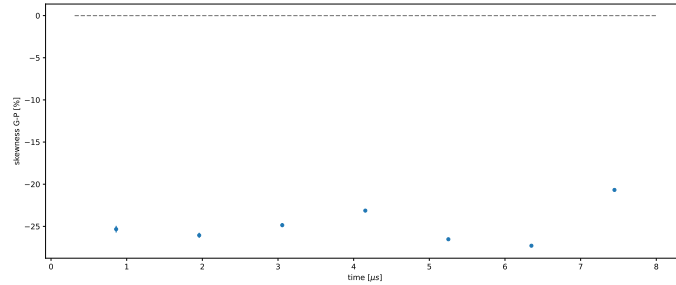


Figure 56: parity violation analysis: new Up-Down asymmetry

We see how with 7 bins the asymmetry is about constant at 25%. We therefore subtract this asymmetry to the asymmetry of the dataset with magnetic field.

We already talked about the model to use in case of parity violation in Eq. (1.16):

$$f_{pv} = A \cos(\omega t + \varphi) + c \quad (5.3)$$

While, in case of parity conservation, the model is a constant line:

$$f_{pc} = c \quad (5.4)$$

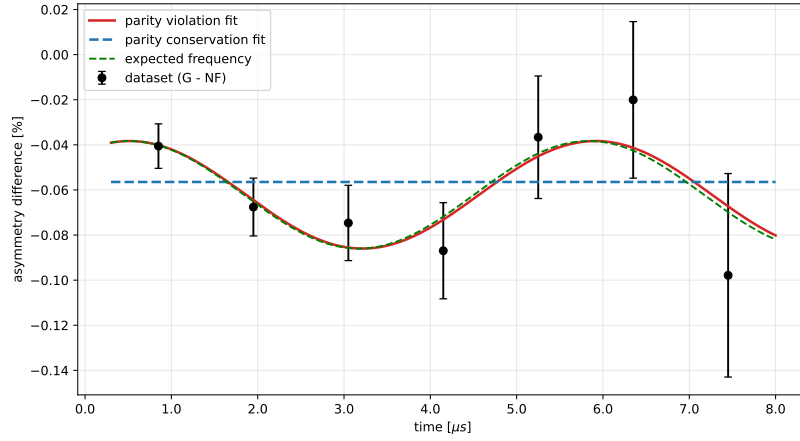


Figure 57: Data asymmetry with both models of parity violation and conservation

Parameter	Value	Uncertainty
A	2.4%	0.9%
φ	0.6 rad	1.1 rad
ω	1.16 rad/ μs	0.26 rad/ μs
c	-6.2%	0.94%

Table 10: Parity violation fit parameters

Parameter	Value	Uncertainty
c	-5.6%	0.63%

Table 11: Parity conservation fit parameters

In Figure 57, we see our dataset compared with the theoretical expectation of parity violation given by the Standard Model and both fits. To confirm the fact that our data show parity violation, we can use the *LogLikelihood-ratio* test to do hypothesis testing.

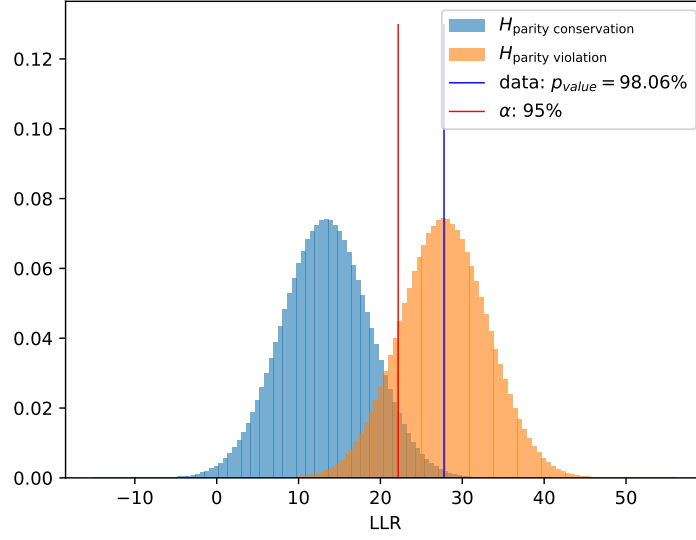


Figure 58: Parity violation: hypothesis testing using the log-likelihood ratio test

We simulated data given the known errors of each point and generate the distribution of the likelihood ratio given the different models. What we found is depicted in Figure 58: we see how our model can reject the null hypothesis with 98.06% probability and therefore we can say that we are able to detect parity violation in our data sample.

Moreover, since the fit rejects the null hypothesis so well, we can study its pulsation value in comparison to what is expected:

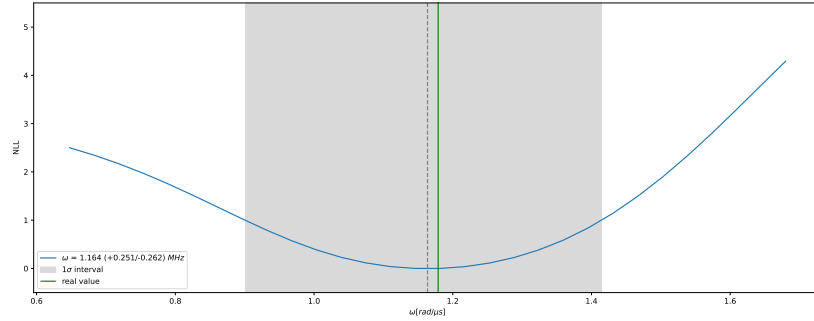


Figure 59: Parity violation: hypothesis testing using the log-likelihood ratio test

We can see how the parameter space of the pulsation is not strictly symmetrical. Although this can seem an issue we can use the upper error to still calculate the goodness of our parameter estimate.

6 Conclusions

With this experiment we were able to obtain decay time values of the muon in the 3 studied materials compatible with the expected value, calculated in Sec. 1.5. For vacuum measurement we obtained a $\tau_{void} = (2.193 \pm 0.011)\mu s$, that gives us a p-value $p = 71.72\%$. For salt measurements we obtained $\tau_{NaCl} = (700 \pm 75)ns$, giving a p-value $p = 94.15\%$. Finally, for aluminum we found $\tau_{Al} = (870 \pm 84)ns$, obtaining a p-value $p = 98.06\%$.

Subsequently we changed our setup in order to have a magnetic field, and measure the up-down asymmetry as a function of the time. We are able to reject the Parity conservation hypothesis with a 98.06% C.L.. As further confirmation, we obtained a value of the pulsation $\omega = (1.16 \pm 0.26)\frac{rad}{\mu s}$, that is compatible with the expected value obtained known the intensity of the magnetic field, with a p-value $p = 95.33\%$. Therefore we can confirm the presence of Parity violation.

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